

Setting the field

This large chapter explores the context of educational research. It sets out several foundations on which different kinds of empirical research are constructed:

- the search for understanding
- paradigms of educational research
- scientific and positivistic methodologies
- naturalistic and interpretive methodologies
- post-positivism, post-structuralism and postmodernism
- complexity theory in educational research

Educational researchers cannot simply ‘read off’ the planning and conduct of research as though one were reading a recipe for baking a cake. Nor is the planning and conduct of research the laboratory world or the field study of the natural scientist. Rather, it is to some degree an art, an iterative and often negotiated process and one in which there are typically trade-offs between what one would like to do and what is actually possible. This book is built on that basis: educational research, far from being a mechanistic exercise, is a deliberative, complex, subtle, challenging, thoughtful activity and often a messier process than researchers would like it to be. This book provides some tools for such deliberation and planning, and hopefully some answers, but beyond that it is for the researcher to consider how to approach, plan, conduct, validate and evaluate the research, how to develop and test theory, how to study and investigate educational matters, how to balance competing demands on the research, and so on. There is no one best way to plan and conduct research, just as there is no one single ‘truth’ to be discovered. Life is not that easy, unidimensional or straightforwardly understood, just as there are no simple dichotomies in educational research (e.g. quantitative *or* qualitative, objective *or* subjective). Rather, we live in a pluralistic world with many purposes and kinds of research, many realities and lived experiences to catch, many outcomes, theories and explanations, many discoveries to be made, and many considerations and often contradictions or sensitivities to be addressed in the planning and conduct of the research.

Whilst arguing against simple foundationalism, this chapter sets out some conceptions of research which researchers may find helpful in characterizing and deliberating about their studies. The chapter considers paradigms and their possible contribution to educational research, positivism, post-positivism, post-structuralism, postmodernism and interpretive approaches.

1.1 Introduction

Our analysis takes an important notion from Hitchcock and Hughes (1995, p. 21), who suggest that ontological assumptions (assumptions about the nature of reality and the nature of things) give rise to epistemological assumptions (ways of researching and enquiring into the nature of reality and the nature of things); these, in turn, give rise to methodological considerations; and these, in turn, give rise to issues of instrumentation and data collection. Added to ontology and epistemology is axiology (the values and beliefs that we hold). This view moves us beyond regarding research methods as simply a technical exercise to being concerned with understanding the world; this is informed by how we view our world(s), what we take understanding to be, what we see as the purposes of understanding and what is deemed valuable.

1.2 The search for understanding

People have long been concerned to come to grips with their environment and to understand the nature of the phenomena it presents to their senses. The means by which they set out to achieve these ends may be classified into three broad categories: *experience*, *reasoning* and *research* (Mouly, 1978). Far from being independent and mutually exclusive, however, these categories are complementary and overlapping, features most readily in evidence where solutions to complex problems are sought.

In our endeavours to come to terms with day-to-day living, we are heavily dependent upon experience and authority. However, as tools for uncovering ultimate truth, they have limitations. The limitations of personal

experience in the form of common-sense knowing, for instance, can quickly be exposed when compared with features of the scientific approach to problem solving. Consider, for example, the striking differences in the way in which theories are used. Laypeople base them on haphazard events and use them in a loose and uncritical manner. When they are required to test them, they do so in a selective fashion, often choosing only that evidence which is consistent with their hunches and ignoring that which is counter to them. Scientists, by contrast, construct their theories carefully and systematically. Whatever hypotheses they formulate have to be tested empirically so that their explanations have a firm basis in fact. And there is the concept of control distinguishing the layperson's and the scientist's attitude to experience. Laypeople may make little or no attempt to control any extraneous sources of influence when trying to explain an occurrence. Scientists, on the other hand, only too conscious of the multiplicity of causes for a given occurrence, adopt definite techniques and procedures to isolate and test the effect of one or more of the alleged causes. Finally, there is the difference of attitude to the relationships among phenomena. Laypeople's concerns with such relationships may be loose, unsystematic and uncontrolled; the chance occurrence of two events in close proximity is sufficient reason to predicate a causal link between them. Scientists, however, display a much more serious professional concern with relationships and only as a result of rigorous experimentation, investigation and testing will they postulate a relationship between two phenomena.

People attempt to comprehend the world around them by using three types of reasoning: *deductive reasoning*, *inductive reasoning* and the *combined inductive-deductive* approach. Deductive reasoning is based on the syllogism, which was Aristotle's great contribution to formal logic. In its simplest form the syllogism consists of a major premise based on an *a priori* or self-evident proposition, a minor premise providing a particular instance, and a conclusion. Thus:

All planets orbit the sun;
The earth is a planet;
Therefore the earth orbits the sun.

The assumption underlying the syllogism is that through a sequence of formal steps of logic, from the general to the particular, a valid conclusion can be deduced from a valid premise. Its chief limitation is that it can handle only certain kinds of statement. The syllogism formed the basis of systematic reasoning from the time of its inception until the Renaissance. Thereafter its effectiveness was diminished because it was no longer related to

observation and experience and became merely a mental exercise. One of the consequences of this was that empirical evidence as the basis of proof was superseded by authority and the more authorities one could quote, the stronger one's position became.

The history of reasoning was to undergo a dramatic change in the 1600s when Francis Bacon began to lay increasing stress on the observational basis of science. Being critical of the model of deductive reasoning on the grounds that its major premises were often preconceived notions which inevitably bias the conclusions, he proposed in its place the method of inductive reasoning by means of which the study of a number of individual cases would lead to a hypothesis and eventually to a generalization. Mouly (1978) explains it by suggesting that Bacon's basic premise was that, with sufficient data, even if one does not have a preconceived idea of their significance or meaning, nevertheless important relationships and laws will be discovered by the alert observer.

Of course, there are limits to induction as the accumulation of a series of examples does not *prove* a theory; it only *supports* it. Just because all the swans that I have ever seen are white, it does not prove a theory that all swans are white – one day I might come across a black swan, and my theory is destroyed. Induction places limits on prediction. Discoveries of associations of regularities and frequent repetitions may have limited predictive value. We are reminded of Bertrand Russell's (1959) story of the chicken who observed that he was fed each day by the same man, and, because this had happened every day, it would continue to happen, i.e. the chicken had a theory of being fed, but, as Russell remarks, 'the man who has fed the chicken every day throughout its life at last wrings its neck instead' (p. 35), indicating the limits of prediction based on observation. Or, to put it more formally, theory is underdetermined by empirical evidence (Phillips and Burbules, 2000, p. 17). Indeed Popper (1980) notes that the essence of science, what makes a science a science, is the inherent falsifiability of the propositions (in contrast to the views of the method of science as being one of verifiability, as held by logical positivists).

This is not to discard induction: it is often the starting point for science. Rather, it is to caution against assuming that it 'proves' anything. Bacon's major contribution to science was that he was able to rescue it from the stranglehold of the deductive method whose abuse had brought scientific progress to a standstill. He thus directed the attention of scientists to nature for solutions to people's problems, demanding empirical evidence for verification. Logic and authority in themselves were no longer regarded as conclusive means of

proof and instead became sources of hypotheses about the world and its phenomena.

Bacon's inductive method was eventually followed by the inductive-deductive approach which combines Aristotelian deduction with Baconian induction. Here the researcher is involved in a back-and-forth process of induction (from observation to hypothesis, from the specific to the general) and deduction (from hypothesis to implications) (Mouly, 1978). Hypotheses are tested rigorously and, if necessary, revised.

Although both deduction and induction have their weaknesses, their contributions to the development of science are enormous, for example: (1) the suggestion of hypotheses; (2) the logical development of these hypotheses; and (3) the clarification and interpretation of scientific findings and their synthesis into a conceptual framework.

A further means by which we set out to discover truth is *research*. This has been defined by Kerlinger (1970) as the systematic, controlled, empirical and critical investigation of hypothetical propositions about the presumed relations among natural phenomena. Research has three characteristics in particular, which distinguish it from the first means of problem solving identified earlier, namely, experience. First, whereas experience deals with events occurring in a haphazard manner, research is systematic and controlled, basing its operations on the inductive-deductive model outlined above. Second, research is empirical. The scientist turns to experience for validation. As Kerlinger puts it, subjective, personal belief must have a reality check against objective, empirical facts and tests. And third, research is self-correcting. Not only does the scientific method have built-in mechanisms to protect scientists from error as far as is humanly possible, but also their procedures and results are open to public scrutiny by fellow professionals. Incorrect results in time will be found and either revised or discarded (Mouly, 1978). Research is a combination of both experience and reasoning and, as far as the natural sciences are concerned, is to be regarded as the most successful approach to the discovery of truth (Borg, 1963).¹

1.3 Conceptions of social reality

The views of social science that we have mentioned represent strikingly different ways of looking at social reality and are constructed on correspondingly different ways of interpreting it. We can perhaps most profitably approach these conceptions of the social world by examining the explicit and implicit assumptions underpinning them. Our analysis is based on the work of Burrell and Morgan (1979), who identified four sets of such assumptions.

First, there are assumptions of an ontological kind – assumptions which concern the very nature or essence of the social phenomena being investigated. Thus, the authors ask, is social reality external to individuals – imposing itself on their consciousness from without – or is it the product of individual consciousness? Is reality of an objective nature, or the result of individual cognition? Is it a given 'out there' in the world, or is it created by one's own mind? Is there a world which exists independent of the individual and which the researcher can observe, discovering relationships, regularities, causal explanations, and which can be tested empirically and repeatedly (i.e. under similar conditions) (cf. Pring, 2015, p. 64)? These questions spring directly from what philosophy terms the nominalist–realist debate. The former view holds that objects of thought are merely words and that there is no independently accessible thing constituting the meaning of a word. The realist position, however, contends that objects have an independent existence and are not dependent for it on the knower. The fact that I can see a dog is not simply because of my perception or cognition but because a dog exists independent of me.

The second set of assumptions identified by Burrell and Morgan are of an epistemological kind. These concern the very bases of knowledge – its nature and forms, how it can be acquired and how communicated to other human beings. How one aligns oneself in this particular debate profoundly affects how one will go about uncovering knowledge of social behaviour. The view that knowledge is hard, objective and tangible will demand of researchers an observer role, together with an allegiance to the methods of natural science; to see knowledge as personal, subjective and unique, however, imposes on researchers an involvement with their subjects and a rejection of the ways of the natural scientist. To subscribe to the former is to be positivist; to the latter, anti-positivist or post-positivist.

The third set of assumptions concern human nature and, in particular, the relationship between human beings and their environment. Since the human being is both its subject and object of study, the consequences for social science of assumptions of this kind are far-reaching. Two images of human beings emerge from such assumptions – the one portrays them as responding mechanically and deterministically to their environment, i.e. as products of the environment, controlled like puppets; the other, as initiators of their own actions with free will and creativity, producing their own environments. The difference is between *determinism* and *voluntarism* respectively (Burrell and Morgan, 1979), between *structure* and *agency*. Human action involves

some combination of these two, polarized here for the sake of conceptual clarity.

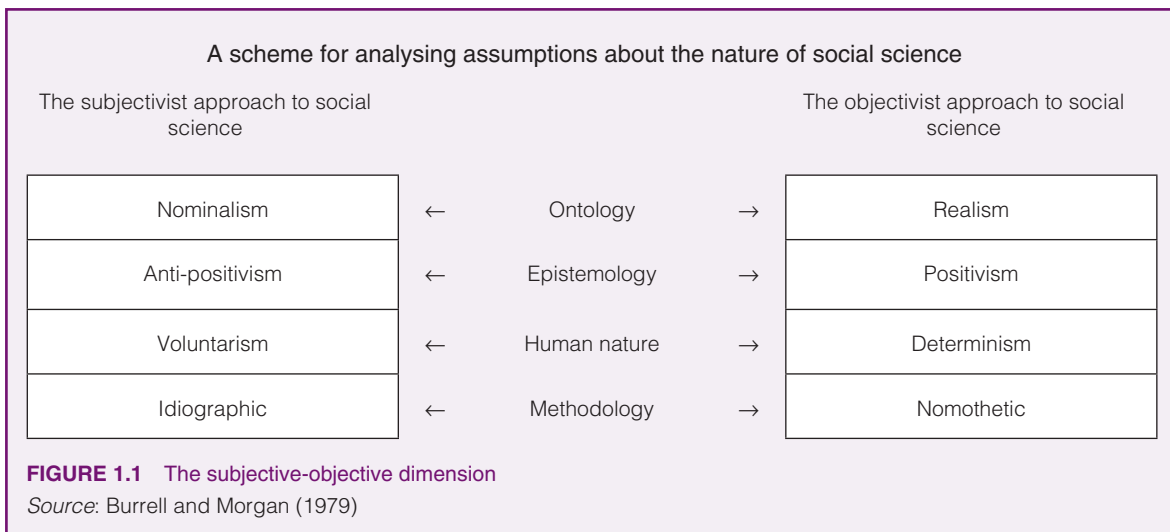
It follows from what we have said so far that the three sets of assumptions identified above have direct implications for the methodological concerns of researchers, since the contrasting ontologies, epistemologies and models of human beings will, in turn, suggest different research methods. Investigators adopting an objectivist (or positivist) approach to the social world and who treat it like the world of natural phenomena as being real and external to the individual will choose from a range of options such as surveys, experiments and the like. Others favouring the more subjectivist (or anti-positivist) approach and who view the social world as being of a much more personal and humanly created kind will select from a comparable range of recent and emerging techniques – accounts, participant observation, interpretive approaches and personal constructs, for example.

Where one subscribes to the view which treats the social world like the natural world – as if it were an external and objective reality – then scientific investigation will be directed at analysing the relationships and regularities between selected factors in that world. It will be concerned with identifying and defining elements and discovering ways in which their relationships can be expressed. Hence, methodological issues, of fundamental importance, are thus the concepts themselves, their measurement and the identification of underlying themes in a search for universal laws which explain and govern that which is being observed (Burrell and Morgan, 1979). An approach characterized by procedures and methods designed to discover general laws may be referred to as *nomothetic*. Here is not the place

to debate whether social life is ‘law-like’ (i.e. can be explained by universal laws) in the same way as that mooted in the natural sciences (but see Kincaid, 2004) or whether social life is quintessentially different from the natural sciences such that ‘law-like’ accounts are simply a search for the impossible and untenable.

However, if one favours the alternative view of social reality which stresses the importance of the subjective experience of individuals in the creation of the social world, then the search for understanding focuses upon different issues and approaches them in different ways. The principal concern is with an *understanding* of the way in which individuals and social groups create, modify and interpret the world in which they find themselves. As Burrell and Morgan (1979) observe, emphasis here is placed on explanation and understanding of the unique and the particular individual cases (however defined: see Chapter 19 on case study, in which emphasis is placed on the denotation of what is the case: an individual, a group, a class, an institution etc.) rather than the general and the universal. In its emphasis on the particular and individual case, this approach to understanding individual (however defined) behaviour may be termed *idiographic*.

In this review of Burrell and Morgan’s analysis of the ontological, epistemological, human and methodological assumptions underlying two ways of conceiving social reality, we have laid the foundations for a more extended study of the two contrasting perspectives evident in the practices of researchers investigating human behaviour and, by adoption, educational problems. Figure 1.1 summarizes these assumptions along a subjective/objective dimension. It identifies the four



sets of assumptions by using terms we have adopted in the text and by which they are known in the literature of social philosophy.

Each of the two perspectives on the study of human behaviour outlined above has profound implications for research in classrooms and schools. The choice of problem, the formulation of questions to be answered,

the characterization of students and teachers, methodological concerns, the kinds of data sought and their mode of treatment, all are influenced by the viewpoint held. Some idea of the considerable practical implications of the contrasting views can be gained by examining Table 1.1, which compares them with respect to a number of critical issues within a broadly societal and

TABLE 1.1 ALTERNATIVE BASES FOR INTERPRETING SOCIAL REALITY

<i>Dimensions of comparison</i>	<i>Conceptions of social reality</i>	
	<i>Objectivist</i>	<i>Subjectivist</i>
Philosophical basis	Realism: the world exists and is knowable as it really is. Organizations are real entities with a life of their own.	Idealism: the world exists but different people construe it in very different ways. Organizations are invented social reality.
The role of social science	Discovering the universal laws of society and human conduct within it.	Discovering how different people interpret the world in which they live.
Basic units of social reality	The collectivity: society or organizations.	Individuals acting singly or together.
Methods of understanding	Identifying conditions or relationships which permit the collectivity to exist. Conceiving what these conditions and relationships are.	Interpretation of the subjective meanings which individuals place upon their action. Discovering the subjective rules for such action.
Theory	A rational edifice built by scientists to explain human behaviour.	Sets of meanings which people use to make sense of their world and behaviour within it.
Research	Experimental or quasi-experimental validation of theory.	The search for meaningful relationships and the discovery of their consequences for action.
Methodology	Abstraction of reality, especially through mathematical models and quantitative analysis.	The representation of reality for purposes of comparison. Analysis of language and meaning.
Society	Ordered. Governed by a uniform set of values and made possible only by those values.	Conflicted. Governed by the values of people with access to power.
Organizations	Goal oriented. Independent of people. Instruments of order in society serving both society and the individual.	Dependent upon people and their goals. Instruments of power which some people control and can use to attain ends which seem good to them.
Organizational pathologies	Organizations get out of kilter with social values and individual needs.	Given diverse human ends, there is always conflict among people acting to pursue them.
Prescription for change	Change the structure of the organization to meet social values and individual needs.	Find out what values are embodied in organizational action and whose they are. Change the people or change their values if you can.

Source: Adapted from Barr Greenfield (1975)

organizational framework. Implications of the two perspectives for educational research unfolds in the course of the text.

1.4 Paradigms

Educational research has absorbed several competing views of the social sciences – the scientific view and an interpretive view – and several others that we explore in this book, including critical theory and feminist theory. Some views hold that the social sciences are essentially the same as the natural sciences and are therefore concerned with discovering natural and universal laws regulating and determining individual and social behaviour. The interpretive view, however, while sharing the rigour of the natural sciences and the concern of social science to describe and explain human behaviour, emphasizes how people differ from inanimate natural phenomena and, indeed, from each other. These contending views – and also their corresponding reflections in educational research – stem in the first instance from different conceptions of social realities and of individual and social behaviour. We examine these in a little more detail.

Since the groundbreaking work of Kuhn (1962), approaches to methodology in research have been informed by discussions of ‘paradigms’ and communities of scholars. A paradigm is a way of looking at or researching phenomena, a world view, a view of what counts as accepted or correct scientific knowledge or way of working, an ‘accepted model or pattern’ (Kuhn, 1962, p. 23), a shared belief system or set of principles, the identity of a research community, a way of pursuing knowledge, consensus on what problems are to be investigated and how to investigate them, typical solutions to problems, and an understanding that is more acceptable than its rivals.

A notable example of this is the old paradigm that placed the Earth at the centre of the universe, only to be replaced by the Copernican heliocentric model, as evidence and explanation became more persuasive of the new paradigm. Importantly, one has to note that the old orthodoxy retained its value for generations because it was supported by respected and powerful scientists and, indeed, others (witness the attempts made by the Catholic Church to silence Galileo in his advocacy of the heliocentric model of the universe). Another example is where the Newtonian view of the mechanical universe has been replaced by the Einsteinian view of a relativistic, evolving universe. More recently still, the idea of a value-free, neutral, objective, positivist science has been replaced by a post-positivist, critical realist view of science with its hallmarks of conjecture

and refutation (Popper, 1980) and with the ability for falsification being the distinguishing feature of science. Further, social science has recognized the importance of the (subjective) value systems of researchers, phenomenology, subjectivity, the need for reflexivity in research (discussed later in this book), the value of qualitative and mixed methods approaches to research, and the contribution of critical theory and feminist approaches to research methodologies and principles.

Paradigms are not simply methodologies (Hammersley, 2013, p. 15); they are ways of looking at the world, different assumptions about what the world is like and how we can understand or know about it. This raises the question of whether paradigms can live together, whether they are compatible or, since they constitute fundamentally different ways of looking at the world, they are incommensurate (which raises questions for mixed methods research – see Chapter 2). At issue here is the significance of regarding approaches to research as underpinned by different paradigms, an important characteristic of which is their incommensurability with each other (i.e. one cannot hold two distinct paradigms simultaneously as there are no common principles, standards or measures).

As more knowledge is acquired to challenge an existing paradigm, such that the original paradigm cannot explain a phenomenon as well as the new paradigm, there comes about a ‘scientific revolution’, a paradigm shift, in which the new paradigm replaces the old as the orthodoxy – the ‘normal science’ – of the day. Kuhn’s (1962) notions of paradigms and paradigm shifts link here objects of study and communities of scholars, where the field of knowledge or paradigm is seen to be only as good as the evidence and the respect in which it is held by ‘authorities’.

Part 1 sets out several paradigms of educational research and these are introduced in Chapters 1 to 3.

Social science research is marked by paradigmatic pluralism and multiple ways of construing paradigms. For example, Pring (2015) contrasts two paradigms (pp. 63–74). The first paradigm espouses the view that there is an objective reality which exists independent of the individual and comprises causally interacting elements which are available for observation; that different sciences (e.g. social, physical) can be used to define that reality once consensus has been reached on what that objective reality is; that the research is replicable and cumulative, i.e. a scientifically rooted body of knowledge can be gathered and checked for correspondence to the world as it is (the correspondence theory of truth) (pp. 63–4). Such a view resonates with Hammersley’s (2013) summary of quantitative research which is characterized by hypothesis testing, numerical

data, ‘procedural objectivity’, generalization, the identification of ‘systematic patterns of association’ and the isolation and control of variables (pp. 10–11).

The second paradigm, by contrast, espouses the view that the world consists of ideas, i.e. a social construction, and that researchers are part of the world which they are researching, that meanings are negotiated between participants (including the researcher), that an objective test of truth is replaced by a consensus theory of truth, that ideas of the world do not exist independently of those who hold them (i.e. require a redefinition of ‘objective’ and ‘subjective’), that multiple realities exist and that what is being researched is context-specific (Pring, 2015, pp. 65–6). Such a view accords with Hammersley’s definition of qualitative research as that which uses less structured data, which emphasizes the central place of subjectivity in the research process and which studies ‘a small number of naturally occurring cases in detail’ using verbal rather than statistical analysis (Hammersley, 2013, p. 12).

However, Pring’s (2015) point is not simply to set out these two paradigms, but to argue that they constitute a false dualism that should be rejected, as they artificially compel the researcher to make an either/or choice of paradigms and, thereby, misrepresent the world as multiply meaningful and both independent of and part of the researcher, not only a social construction. He argues (p. 69) that, just as an independent physical world must exist in order for researchers to construe it, the same can be said of the social world – there must be independent actors and social worlds in order for apperception and social construction of it to make sense.

Pring cautions against adopting *a priori* either a quantitative or qualitative view of the world as this massively over-simplifies the real world, which is complex and complicated. Rather, how we pursue the research depends on what the research is about, and this recognizes that social constructions vary from social group to social group and humans can be both the object and subject of research (2015, p. 73).

Pring is not alone in characterizing different paradigms of educational research. For example, Creswell (2013) notes four ‘philosophical worldviews’ (pp. 7ff.): post-positivism, constructivism, advocacy/participatory and pragmatism. These are discussed in Chapters 2 and 3. Here we note that the advocacy/participatory paradigm concerns the disempowered and marginalized, and it studies oppression and lack of voice; this brings it under the umbrella of critical approaches which we discuss in Chapter 3, including gender, race, ethnicity, disability, sexual orientation, socio-economic status and differentials of power that prop up inequality.

Lather (2004) sets out four paradigms: prediction (positivism); understanding (interpretive approaches); emancipatory (critical theoretical approaches); and deconstruction (post-structuralist). We discuss these in Chapters 1 to 3. Lukenchuk (2013) identifies six paradigms which, she notes, are not exhaustive (pp. 66ff.):

- Empirical-analytic (empiricist; scientific; concerned with prediction and control; quantitative; experimental; correlational; causal; explanatory; probabilistic; fallibilistic; concerned with warrants for knowledge claims; quantitative);
- Pragmatic (focus on ‘what works’; trial and error; problem-centred; practical; experimental; action oriented; utility oriented; practitioner research; qualitative and quantitative);
- Interpretive (hermeneutic and existential understanding; meaning-making; phenomenological; qualitative; naturalistic; constructivist; interactionist; verstehen approaches; ethnographic; qualitative);
- Critical (ideology-critical; concerned with analysis of power and ideology; consciousness-raising; emancipatory and concerned with advocacy/participatory approaches; transformatory; politically oriented and activist; qualitative and quantitative);
- Post-structuralist (anti-foundation knowledge; deconstructionist; interpretation of life as discourse and texts; transformative; qualitative);
- Transcendental (asserts reason, intuition, mysticism, revelation as ways of knowing: mind, body, soul and spirit; life as directed by an ‘internal moral compass’; foundational; qualitative).

This is not to say that paradigms necessarily *drive* the research, as research is driven by the purposes of the research. Indeed we can ask whether we need paradigmatic thinking at all in order to do research. Rather, it is to say that the purposes and nature of the research may be clarified by drawing on one or more of these paradigms; the paradigms can clarify and organize the thinking about the research. Further, it is not to say that these paradigms each have an undisputed coherence, unity or unproblematic singularity of conception. Rather, they are characterizations, ideal types, typifications and simplifications for ease of initial understanding, recognizing that this blurs the many variations that lie within each of them, and, indeed, may overlook the overlaps between them; each paradigm is not all of a single type and they are by no means mutually exclusive. To consider them as mutually exclusive is to prolong the unnecessary ‘paradigm wars’ to which Gage (1989) alluded so compellingly.

Because of its significance for the epistemological basis of social science and its consequences for educational research, we devote discussion in this chapter to the debate on positivism and anti-positivism/post-positivism, and on alternative paradigms and rationales for understanding educational research.

1.5 Positivism

Although positivism has been a recurrent theme in the history of western thought from the Ancient Greeks to the present, it is historically associated with the nineteenth-century French philosopher, Auguste Comte, who was the first thinker to use the word for a philosophical position (Beck, 1979) and who gave rise to sociology as a distinct discipline. His positivism turns to observation and reason as means of understanding behaviour, i.e. empirical observation and verification; explanation proceeds by way of scientific description. In his study of the history of the philosophy and methodology of science, Oldroyd (1986) says that, in this view, social phenomena could be researched in ways similar to natural, physical phenomena, i.e. generating laws and theories that could be investigated empirically.

Comte's position was to lead to a general doctrine of positivism which held that all genuine knowledge is based on sensory experience and can only be advanced by means of observation and experiment: the scientific method. Following in the empiricist tradition, it limited enquiry and belief to what can be firmly established and in thus abandoning metaphysical and speculative attempts to gain knowledge by reason alone, the movement developed a rigorous orientation to social facts and natural phenomena to be investigated empirically (Beck, 1979). Taking account of this, matters of values were out of court for the positivist, as they were not susceptible to observation evidence, i.e. there is a separation between facts and values.

With its emphasis on observational evidence and the scientific method, positivism accords significance to sensory experience (empiricism), observational description (e.g. ruling our inferences about actors' intentions, thoughts or attitudes), operationalism, 'methodical control', measurement, hypothesis testing and replicability through the specification of explicit and transparent procedures for conducting research (Hammersley, 2013, pp. 23–4). Hammersley notes that the terms 'positivism' and 'empiricism' are often regarded as synonymous with each other (p. 23), but to equate positivism simply with quantitative approaches is misguided, as qualitative data are equally well embraced within empiricism. Indeed he notes that ethnographers and

discourse analysts rely on careful observational data (pp. 24–5).

Though the term positivism is used by philosophers and social scientists, a residual meaning derives from an acceptance of natural science as the paradigm of human knowledge (Duncan, 1968). This includes the following connected suppositions, identified by Giddens (1975). First, the methodological procedures of natural science may be directly applied to the social sciences. Positivism here implies a particular stance concerning the social scientist as an observer of social reality. Second, the end-product of investigations by social scientists can be formulated in terms parallel to those of natural science. This means that their analyses must be expressed in laws or law-like generalizations of the same kind that have been established in relation to natural phenomena. Positivism claims that science provides us with the clearest possible ideal of knowledge.

Where positivism is less successful, however, is in its application to the study of human behaviour, where the immense complexity of human nature and the elusive and intangible quality of social phenomena contrast strikingly with the order and regularity of the natural world. This point is apparent in the contexts of classrooms and schools where the problems of teaching, learning and human interaction present the positivistic researcher with a mammoth challenge.

We now look more closely at some of the features of the scientific method that is underpinned by positivism.

1.6 The assumptions and nature of science

We begin with an examination of the tenets of scientific faith: the kinds of assumptions held by scientists, often implicitly, as they go about their daily work. First, there is the assumption of *determinism*. This means simply that events have causes; that events are determined by other circumstances; and science proceeds on the belief that these causal links can eventually be uncovered and understood. Moreover, not only are events in the natural world determined by other circumstances, but there is regularity about the way in which they are determined: the universe does not behave capriciously. It is the ultimate aim of scientists to formulate laws to account for the happenings in the world, thus giving them a firm basis for prediction and control.

The second assumption is that of *empiricism*, which holds that certain kinds of reliable knowledge can only derive from experience. This is an example of foundationalism. In this case, to quote the philosopher John

Locke (1959): ‘whence has it [the mind] all the materials of reason and knowledge? To this, I answer, in one word, from experience. In that all knowledge is founded; and from that it ultimately derives itself’ (p. 26). Experience means sensory experience, and this contrasts with the rationalist epistemology in which reason rules supreme. In empiricism, experience alone provides the warrant for, or justification of, a knowledge claim, which is brought to the scientific community for acceptance. Such empiricism gives rise to the need for the operationalization of concepts, for example, creativity, intelligence, ability (Phillips and Burbules, 2000, p. 10), in order for them to be observable. Empiricism (and positivism) does not preclude non-experimental studies, nor does it prescribe only quantitative research.

In practice, empiricism means scientifically that the tenability of a theory or hypothesis depends on the nature of the empirical evidence for its support. ‘Empirical’ here means that which is verifiable by observation, direct experience and evidence, data-yielding proof or strong confirmation, in probability terms, of a theory or hypothesis in a research setting.

Mouly (1978) identifies five steps in the process of empirical science:

- 1 *Experience* – the starting point of scientific endeavour at the most elementary level;
- 2 *Classification* – the formal systematization of otherwise incomprehensible masses of data;
- 3 *Quantification* – a more sophisticated stage where precision of measurement allows more adequate analysis of phenomena by mathematical means;
- 4 *Discovery of relationships* – the identification and classification of functional relationships among phenomena;
- 5 *Approximation to the truth* – science proceeds by gradual approximation to the truth.

The third assumption underlying the work of the scientist is the principle of *parsimony*. The basic idea is that phenomena should be explained in the most economical way possible. As Einstein was known to remark, one should make matters as simple as possible, but no simpler! The first historical statement of the principle was by William of Occam when he said that explanatory principles (entities) should not be needlessly multiplied (‘Occam’s razor’), i.e. that it is preferable to account for a phenomenon by two concepts rather than three; that a simple theory is to be preferred to a complex one.

The final assumption, that of *generality*, played an important part in both the deductive and inductive methods of reasoning. Indeed, historically speaking, it was the problematic relationship between the concrete particular and the abstract general that was to result in two competing theories of knowledge – the rational and the empirical. Beginning with observations of the particular, scientists set out to generalize their findings to the world at large. This is because they are concerned ultimately with explanation. Of course, the concept of generality presents much less of a problem to natural scientists working chiefly with inanimate matter than to human scientists who, of necessity having to deal with samples of larger human populations, must exercise great caution when generalizing their findings to the particular parent populations.

We come now to the core question: What is science? Kerlinger (1970) points out that in the scientific world itself two broad views of science may be found: the *static* and the *dynamic*. The *static* view, which has particular appeal for laypeople, is that science is an activity that contributes systematized information to the world. The work of the scientist is to uncover new facts and add them to the existing corpus of knowledge. Science is thus seen as an accumulated body of

BOX 1.1 THE FUNCTIONS OF SCIENCE

- 1 Its problem-seeking, question-asking, hunch-encouraging, hypotheses-producing function.
- 2 Its testing, checking, certifying function; its trying out and testing of hypotheses; its repetition and checking of experiments; its piling up of facts.
- 3 Its organizing, theorizing, structuring function; its search for larger and larger generalizations.
- 4 Its history-collecting, scholarly function.
- 5 Its technological side; instruments, methods, techniques.
- 6 Its administrative, executive and organizational side.
- 7 Its publicizing and educational functions.
- 8 Its applications to human use.
- 9 Its appreciation, enjoyment, celebration and glorification.

Source: Maslow (1954)

findings, the emphasis being chiefly on the present state of knowledge and adding to it.² The *dynamic* view, by contrast, conceives science more as an activity, as something that scientists *do*. According to this conception it is important to have an accumulated body of knowledge of course, but what really matter most are the discoveries that scientists make. The emphasis here, then, is more on the heuristic nature of science.

Contrasting views exist on the functions of science. We give a composite summary of these in Box 1.1. For professional scientists, however, science is seen as a way of comprehending the world; as a means of explanation and understanding, of prediction and control. For them the ultimate aim of science is theory, and we discuss this in Chapter 4.

We look now in more detail at two such tools which play a crucial role in science – the concept and the hypothesis.

1.7 The tools of science

Concepts express generalizations from particulars – anger, achievement, alienation, velocity, intelligence, democracy. Examining these examples more closely, we see that each is a word representing an idea: more accurately, a concept is the relationship between the word (or symbol) and an idea or conception. Whoever we are and whatever we do, we all make use of concepts. Naturally, some are shared and used by all groups of people within the same culture – child, love, justice, for example; others, however, have a restricted currency and are used only by certain groups, specialists or members of professions – idioglossia, retroactive inhibition, anticipatory socialization.

Concepts enable us to impose some sort of meaning on the world; through them reality is given sense, order and coherence. They are the means by which we are able to come to terms with our experience. How we perceive the world, then, is highly dependent on the repertoire of concepts that we have. The more we have, the more sense data we can pick up and the surer will be our perceptual (and cognitive) grasp of whatever is ‘out there’. If our perceptions of the world are determined by the concepts available to us, it follows that people with differing sets of concepts will tend to view the ‘same’ objective reality differently – a doctor diagnosing an illness will draw upon a vastly different range of concepts from, say, the restricted and perhaps simplistic notions of the layperson in that context.

So where is all this leading? Simply to this: social scientists have likewise developed, or appropriated by giving precise meaning to, a set of concepts which enable them to shape their perceptions of the world in a

particular way, to represent that slice of reality which is their special study. And collectively, these concepts form part of their wider meaning system which permits them to give accounts of that reality, accounts which are rooted and validated in the direct experience of everyday life, for example, the concept of social class which offers researchers ‘a rule, a grid, even though vague at times, to use in talking about certain sorts of experience that have to do with economic position, life-style, life-chances, and so on’ (Hughes, 1976, p. 34).

There are two important points to stress when considering scientific concepts. The first is that they do not exist independently of us: they are our inventions, enabling us to acquire some understanding of nature. The second is that they are limited in number and in this way contrast with the infinite number of phenomena they are required to explain.

A second tool of great importance to the scientist is the *hypothesis*. It is from this that much research proceeds, especially where cause-and-effect or concomitant relationships are being investigated. The hypothesis has been defined by Kerlinger (1970) as a conjectural statement of the relations between two or more variables, or ‘an educated guess’, though it is unlike an educated guess in that it is often the result of considerable study, reflective thinking and observation. Medawar (1972) writes of the hypothesis and its function as being speculative and imaginative preconceptions or conjectures about what might be true, which are subject to criticism to see if they really are like the phenomenon in question. As he remarks, scientific reasoning is a dialogue between the ‘imaginative and the critical’, the ‘possible and the actual’, between ‘what might be true and what is in fact the case’ (Medawar, 1972, p. 22).

Kerlinger (1970) has identified two criteria for ‘good’ hypotheses. The first is that hypotheses are statements about the relations between variables; and second, that hypotheses carry clear implications for testing the stated relations. To these he adds two ancillary criteria: that hypotheses disclose compatibility with current knowledge; and that they are expressed as economically as possible. Thus if we conjecture that social class background determines academic achievement, we have a relationship between one variable, social class, and another, academic achievement. And since both can be measured, the primary criteria specified by Kerlinger can be met. Neither do they violate the ancillary criteria he proposed (see also Box 1.2).

Kerlinger further identifies four reasons for the importance of hypotheses as tools of research. First, they organize the efforts of researchers. The relationship expressed in the hypothesis indicates what they should do. They enable them to understand the problem

BOX 1.2 THE HYPOTHESIS

Once one has a hypothesis to work on, the scientist can move forward; the hypothesis will guide the researcher on the selection of some observations rather than others and will suggest experiments. Scientists soon learn by experience the characteristics of a good hypothesis. A hypothesis that is so loose as to accommodate *any* phenomenon tells us precisely nothing; the more phenomena it prohibits, the more informative it is.

A good hypothesis must also have *logical immediacy*, i.e. it must provide an explanation of whatever it is that needs to be explained and not an explanation of other phenomena. Logical immediacy in a hypothesis means that it can be tested by comparatively direct and practicable means. A large part of the *art of the soluble* is the art of devising hypotheses that can be tested by practicable experiments.

Source: Adapted from Medawar (1981)

with greater clarity and provide them with a framework for collecting, analysing and interpreting their data. Second, they are, in Kerlinger's words, the working instruments of theory. They can be deduced from theory or from other hypotheses. Third, they can be tested, empirically or experimentally, resulting in confirmation or rejection. And there is always the possibility that a hypothesis, once supported and established, may become a law. And fourth, hypotheses are powerful tools for the advancement of knowledge because, as Kerlinger explains, they enable us to get outside ourselves. Hypotheses and concepts play a crucial part in the scientific method and it is to this that we now turn our attention.

1.8 The scientific method

If the most distinctive feature of science is its empirical nature, the next most important characteristic is its set of procedures which show not only how findings have been arrived at, but are sufficiently clear for fellow-scientists to repeat them, i.e. to check them out with the same or other materials and thereby test the results. As Cuff and Payne (1979) say: 'A scientific approach

necessarily involves standards and procedures for demonstrating the "empirical warrant" of its findings, showing the match or fit between its statements and what is happening or has happened in the world' (Cuff and Payne, 1979, p. 4). For convenience we will call these standards and procedures 'the scientific method', though this can be somewhat misleading, as the combination of the definite article, adjective and singular noun risks conjuring up a single invariant approach to problem solving. Yet there is much more to it than this. The term in fact cloaks a number of methods which vary in their degree of sophistication depending on their function and the particular stage of development a science has reached.

The scientific method initially involves systematic observation, moving to interconnecting ideas coherently and without internal contradictions (creating a scientific model), which is then tested by further observations (Capra and Luisi, 2014). Box 1.3 sets out the sequence of stages through which a science normally passes in its development or, perhaps more realistically, that are constantly present in its progress and on which scientists may draw depending on the kind of information they seek or the kind of problem confronting them.

BOX 1.3 STAGES IN THE DEVELOPMENT OF A SCIENCE

- 1 Definition of the science and identification of the phenomena that are to be subsumed under it.
- 2 Observational stage at which the relevant factors, variables or items are identified and labelled; and at which categories and taxonomies are developed.
- 3 Correlational research in which variables and parameters are related to one another and information is systematically integrated as theories begin to develop.
- 4 The systematic and controlled manipulation of variables to see if experiments will produce expected results, thus moving from correlation to causality.
- 5 The firm establishment of a body of theory as the outcomes of the earlier stages are accumulated. Depending on the nature of the phenomena under scrutiny, laws may be formulated and systematized.
- 6 The use of the established body of theory in the resolution of problems or as a source of further hypotheses.

Of particular interest in our efforts to elucidate the term ‘scientific method’ are stages 2, 3 and 4. Stage 2 is a relatively uncomplicated point at which the researcher is content to observe and record facts and possibly arrive at some system of classification. Much research in the field of education is conducted in this way, for example, surveys and case studies. Stage 3 establishes relationships between variables within a loose framework of inchoate theory. Stage 4 is the most sophisticated stage and often the one that many people equate exclusively with the scientific method. In order to arrive at causality, as distinct from mere measures of association, researchers here design experimental situations in which variables are manipulated to test their chosen hypotheses. This process moves from early, inchoate ideas, to more rigorous hypotheses, to empirical testing of those hypotheses, thence to confirmation or modification of the hypotheses (Kerlinger, 1970).

Hitchcock and Hughes (1995, p. 23) suggest an eight-stage model of the scientific method that echoes Kerlinger. This is represented in Box 1.4.

The elements the researchers fasten on to will naturally be suitable for scientific formulation; this means simply that they will possess quantitative aspects. Their principal working tool will be the hypothesis which, as we have seen, is a statement indicating a relationship (or its absence) between two or more of the chosen elements and stated in such a way as to carry clear implications for testing. Researchers then choose the most appropriate method and put their hypotheses to the test.

1.9 Criticisms of positivism and the scientific method

In spite of the scientific enterprise’s proven success using positivism – especially in the field of natural science – its ontological and epistemological bases have been the focus of sustained and sometimes vehement criticism from some quarters. Beginning in the second

half of the nineteenth century, the revolt against positivism occurred on a broad front. Essentially, it has been a reaction against the world picture projected by science which, it is contended, undermines life and mind. The precise target of the anti-positivists’ attack has been science’s mechanistic and reductionist view of nature which, by definition, regards life in measurable terms rather than inner experience, and excludes notions of choice, freedom, individuality and moral responsibility, regarding the universe as a living organism rather than as a machine (e.g. Nesfield-Cookson, 1987).

Here the putative objectivity of science is called into question, and objectivity is treated as problematic. Kettley (2012), for example, notes that objective knowledge is often treated as unproblematic and viewed through simplistic, unacceptably reductionist lenses in which empiricism is reduced to knowing through observation, positivism is viewed as Comte’s rebuttal of metaphysics, that there is a unity between the scientific method and Durkheim’s positivism, and realism is a synonym for undisputed existence (p. 71). However, he contends, objective knowledge is actually contested, subjective meanings affect or refract views of what are generally considered to be objective knowledge and objectivity (e.g. social facts) which do not necessarily reside in the phenomenon itself but in the subjective values of the researcher (p. 72), and that equating the scientific methods with positivism overlooks the important distinction between induction and deduction. Douglas (2004) notes that the very term ‘objective’ is fraught with definitional problems, and he gives several senses in which it is used, including, for example: manipulable, detached, procedural, value-neutral and value-free.

The point is well made: objectivity and objective knowledge are beset with problems, and researchers are well advised to avoid simple dichotomies or absolutist ideal types: objective or subjective, induction or deduction, quantitative or qualitative. Rather, there is no unified objectivist or subjectivist paradigm (Kettley,

BOX 1.4 AN EIGHT-STAGE MODEL OF THE SCIENTIFIC METHOD

- Stage 1: Hypotheses, hunches and guesses
- Stage 2: Experiment designed; samples taken; variables isolated
- Stage 3: Correlations observed; patterns identified
- Stage 4: Hypotheses formed to explain regularities
- Stage 5: Explanations and predictions tested; falsifiability
- Stage 6: Laws developed or disconfirmation (hypothesis rejected)
- Stage 7: Generalizations made
- Stage 8: New theories

2012, p. 76); objective reality is constructed subjectively; positivism is not a unified, singular, coherent tenet; hypothesis formation is a human act that derives in part from the subjective views of the researcher (and these subjective views can differ sharply); aggregated data do not override or negate the constructions and meanings accorded to a situation by individuals; and the assumption of linear relationships is frustrated by a non-linear world (pp. 76–7).

Another challenge to the claims of positivism came from Søren Kierkegaard, the Danish philosopher, one of the originators of existentialism. Kierkegaard was concerned with individuals and their need to fulfil themselves to the highest level of development. This realization of a person's potential was for him the meaning of existence which he saw as concrete and individual, unique and irreducible, not amenable to conceptualization (Beck, 1979). Features of the age in which we live – the ascendancy of scientific and technological progress – militate against the achievement of this end and contribute to the dehumanization of the individual. In his desire to free people from their illusions, the illusion Kierkegaard was most concerned about was that of objectivity. By this he meant the imposition of rules of behaviour and thought, and the making of a person into an observer set on discovering general laws governing human behaviour. The capacity for subjectivity, he argued, should be regained and retained. This he regarded as the ability to consider one's own relationship to whatever constitutes the focus of enquiry.

Also concerned with the dehumanizing effects of the social sciences is Ions (1977). While acknowledging that they can take much credit for throwing light in dark corners, he expresses serious concern at the way in which quantification and computation, assisted by statistical theory and method, are used. He argues that quantification is a form of collectivism, but that this runs the risk of depersonalization. His objection is not directed at quantification *per se*, but at quantification when it becomes an end in itself, replacing humane study which seeks to investigate and shed light on the human condition (Ions, 1977). This echoes Horkheimer's (1972) powerful critique of positivism as the mathematization of concepts about nature and of scientism – science's belief in itself as the only way of conducting research and explaining phenomena.

Another forceful critic of the objective consciousness has been Roszak (1970, 1972), who argues that science, in its pursuit of objectivity, is a form of alienation from our true selves and from nature. The justification for any intellectual activity lies in the effect it has on increasing our awareness and degree of consciousness,

but this increase, some claim, has been retarded in our time by the excessive influence that the positivist paradigm has exerted on areas of our intellectual life. Holbrook (1977), for example, affording consciousness a central position in human existence and deeply concerned with what happens to it, condemns positivism and empiricism for their bankruptcy of the inner world, morality and subjectivity.

Hampden-Turner (1970) concludes that the social science view of human beings is a restricted image of humans when social scientists concentrate on the repetitive, predictable and invariant aspects of the person; on 'visible externalities' to the exclusion of the subjective world; and on the parts of the person in their endeavours to understand the whole.

Habermas (1972), in keeping with the Frankfurt School of critical theory (discussed in Chapter 3), provides a corrosive critique of positivism, arguing that the scientific mentality has been elevated to an almost unassailable position – almost to the level of a religion (scientism) – as being the only epistemology of the west. In this view all knowledge becomes equated with scientific knowledge. This neglects hermeneutic, aesthetic, critical, moral, creative and other forms of knowledge. It reduces behaviour to technicism.

Positivism's concern for control and, thereby, its appeal to the passivity of behaviourism and for instrumental reason is a serious danger to the more open-ended, creative, humanitarian aspects of social behaviour. Habermas (1972, 1974) and Horkheimer (1972) argue that scientism silences an important debate about values, informed opinion, moral judgements and beliefs. Scientific explanation seems to be the only means of explaining behaviour, and, for them, this seriously diminishes the very characteristics that make humans human. It makes for a society without conscience. Positivism is unable to answer many interesting or important areas of life (Habermas, 1972, p. 300), resonating with Wittgenstein's (1974) comment that when all possible scientific questions have been addressed, they have left untouched the main problems of life.

Other criticisms are commonly levelled at positivistic social science. One is that it fails to take account of our unique ability to interpret our experiences and represent them to ourselves. How we make sense of the social world resides in our distinctively human nature, and we have to take account of this in recognizing that the social world is not the same as an object of science (Pring, 2015, p. 115) (though Durkheim noted that there are 'social facts', i.e. those that transcend individuals' interpretations and constructions). We can, and do, construct theories about ourselves and our world,

and we act on these theories. In failing to recognize this, positivistic social science is said to ignore the profound differences between itself and the natural sciences. Social science, unlike natural science, stands in a subject–subject rather than a subject–object relation to its field of study, and works in a pre-interpreted world in the sense that the meanings that subjects hold are part of their construction of the world (Giddens, 1976).

The difficulty in which positivism finds itself is that it regards human behaviour as passive, essentially determined and controlled, thereby ignoring intention, individualism and freedom, i.e. as suffering from the same difficulties that inhere in behaviourism (see Chomsky's (1959) withering criticism). This problem with positivism also rehearses the familiar problem in social theory, namely, the tension between agency and structure (Layder, 1994): humans exercise agency – individual choice and intention – not necessarily in circumstances of their own choosing, but nevertheless they do not behave simply or deterministically like puppets.

Finally, the findings of positivistic social science are often said to be so banal and trivial that they are of little consequence to those for whom they are intended, namely, teachers, social workers, counsellors, managers and the like. The more effort, it seems, that researchers put into their scientific experimentation in the laboratory by restricting, simplifying and controlling variables, the more likely they are to end up with a stripped down, artificial, deterministic view of the world as if it were a laboratory.³

These are formidable criticisms; but what alternatives are proposed by the detractors of positivistic social science?

1.10 Post-positivism

The positivist view of the world is of an ordered, controllable, predictable, standardized, mechanistic, deterministic, stable, objective, rational, impersonal, largely inflexible, closed system whose study yields immutable, absolute, universal laws and patterns of behaviour (a 'grand narrative', a 'metanarrative') and which can be studied straightforwardly through the empirical, observational means of the scientific method. It suggests that there are laws of cause and effect, often of a linear nature (a specific cause produces a predictable effect, a small cause (stimulus) produces a small effect (response) and a large cause produces a large effect), which can be understood typically through the application of the scientific method as set out earlier in this chapter. Like a piece of clockwork, there is a place for

everything and everything is in its place. It argues for an external and largely singular view of an objective reality (i.e. external to, and independent of, the researcher) that is susceptible to scientific discovery and laws. However, as Lukenchuk (2013) notes, positivism has been discarded as a useful scientific paradigm as it has failed to provide a 'logically unified system of theoretical statements grounded in the certainty of sense experience' (p. 16) and has been superseded by post-positivism.

Post-positivists challenge the positivist view of the world. Here, following Popper (1968, 1980), our knowledge of the world is not absolute but partial, conjectural, falsifiable, challengeable, provisional, probabilistic and changing. Whilst still embracing the scientific method and the acceptance of an objective world, it recognizes that there is no absolute truth, or, at least, not one which is discoverable by humans, but, rather, probabilistic knowledge only. Secure, once-and-for-all foundational knowledge and grand narratives of a singular objective reality, discoverable through empiricism, positivism, behaviourism and rationalism, are replaced by tentative speculation in which multiple perspectives, claims and warrants are brought forward by the researcher (Phillips and Burbules, 2000). The world is multilayered, able to tolerate multiple interpretations, and in which – depending on the particular view of post-positivism that is being embraced – there exist multiple external realities; knowledge is regarded as subjective rather than objective. In short, the values, biographies, perceptions, theories, environment and existing knowledge of researchers influence what is being observed, and this undermines the foundationalism of empiricism with its claims to neutral sensory experience and observation (Phillips and Burbules, 2000, p. 17). As mentioned earlier, theory is underdetermined by evidence, as the same evidence can support several different theories.

Post-positivists argue that facts and observations are theory-laden and value-laden (Feyerabend, 1975; Popper, 1980; Reichardt and Rallis, 1994), facts and theories are fallible, different theories may support specific observations/facts, and social facts, even ways of thinking and observing, are social constructions rather than objectively and universally true (Nisbett, 2005).

Imagine that a researcher observes a class lesson and notices one student winking at the teacher. Is this student being cheeky (a theory of deviant or challenging behaviour), a sign of understanding (a theory of cognition/recognition), a physical problem (Tourette's syndrome), a sign of stress or happiness (a theory of emotional behaviour), a sign of friendliness (a theory of interpersonal non-verbal behaviour), or what? The

observation on its own cannot tell us. There is a gap between an observed phenomenon and the explanation or theory of, or a hypothesis about, the phenomenon. As Phillips and Burbules (2000, pp. 18–19) remark, phenomena do not speak for themselves. This gap cannot be bridged by observed evidence alone, but needs help from outside that observed phenomenon, i.e. from non-sensory experience. What we see depends on our viewpoint. This is not to say that there is no correct answer or that multiple interpretations are acceptable (relativism), only that the observation alone is not sufficient to denote meaning.

Out goes foundational knowledge and in comes non-foundational, tentative, conjectural speculation and probabilistic, fallibilistic, imperfect, context-bound knowledge of multiple truths of a situation and multiple realities, whose validity has to be warranted whilst recognizing that such warrants may be overturned in light of future evidence. Here the separation of fact and value in positivism is unsustainable, and the foundationalism of empiricism is replaced by an admission that observation is theory-laden, and our values, perspectives, paradigms, conceptual schemes, even research communities determine what we focus on, how we research, what we deem to be important, what counts as knowledge, what research ‘shows’, how we interpret research findings and what constitutes ‘good’ research.

Post-positivism argues for the continuing existence of an objective reality, i.e. it rejects relativism, but it adopts a pluralist view of multiple, coexisting realities rather than a single reality. Imagine that two people are observing a classroom; one sits at the back of the room, and the other at the front. What they see may differ, but it is still the same classroom. Multiple views are not the same as relativism; multiple truths can coexist. There is an objective reality: the classroom, but there are different views of this, i.e. ‘truth’ is not simply what one of the observers takes it to be, and one frame of reference may differ from another. This raises the issue of bias and value-neutrality in educational research, which we discuss in Chapter 3.

Post-positivism recognizes that we know the world only probabilistically and imperfectly. Whilst not rejecting the value of the scientific method (e.g. experimentation), it argues for the reformulation of the strength of theories and claims made from the scientific method, namely, that their strengths are contingent on their ability to withstand ‘severe tests’ of their falsifiability and that their discoveries are subject to future falsification in the light of new evidence. Seen in this light, the gap between natural sciences and social science evaporates. In the post-positivist view of

science, characterized by the theory-laden nature of observations, the underdetermination of theory by empirical evidence, the importance of the community of scholars in validating warrants for knowledge, the tentative, conjectural nature of conclusions, and the multiple nature of reality and ‘truths’, the researcher in the natural sciences is in no more or less a privileged position than the social science researcher.

1.11 Alternatives to positivistic and post-positivist social science: naturalistic and interpretive approaches

Although opponents of positivism within social science subscribe to a variety of schools of thought, each with its own different epistemological viewpoint, they are united by their common rejection of the belief that human behaviour is governed by general, universal laws and characterized by underlying regularities. Moreover, they would agree that the social world can only be understood from the standpoint of the individuals who are part of the ongoing action being investigated and that their model of a person is an autonomous one, not the version favoured by positivist researchers. Such a view is allied to constructivism (Creswell, 2013) and to interpretive approaches to social science (discussed below).

In rejecting the viewpoint of the detached, objective observer – a mandatory feature of traditional research – anti-positivists and post-positivists would argue that individuals’ behaviour can only be understood by the researcher sharing their frame of reference: understanding of individuals’ interpretations of the world around them has to come from the inside, not the outside. Social science is thus seen as a subjective rather than an objective undertaking, as a means of dealing with the direct experience of people in specific contexts, where social scientists understand, explain and demystify social reality through the eyes of different participants; the participants themselves define the social reality (Beck, 1979). This is not to say that understanding subjective meanings is the only route for the researcher. Rather it is both a question of emphasis and a recognition that there are external matters that impinge on subjective meaning-making and, indeed, that what constitutes ‘subjectivity’ is open to question and to multiple interpretations and consequences, rather than being a unified, coherent singularity (Kettley, 2012, pp. 78–9). Subjective meanings may be as empirically testable as objective statements.

The anti-positivist/post-positivist movement has many hues, for example, postmodernism, post-structuralism

and Wittgenstein's work on language games. These have influenced areas of social science such as psychology, social psychology and sociology. In the case of psychology, for instance, a school of humanistic psychology has emerged alongside the coexisting behaviouristic and psychoanalytic schools. Arising as a response to the challenge to combat the growing feelings of dehumanization which characterize many social and cultural milieux, it sets out to study and understand the person as a *whole* (Buhler and Allen, 1972). Humanistic psychologists present a model of people that is positive, active and purposeful, and at the same time stresses their own involvement with the life experience itself. They do not stand apart, introspective, hypothesizing. Their interest is directed at the intentional and creative aspects of the human being. The perspective adopted by humanistic psychologists is naturally reflected in their methodology. They are dedicated to studying the individual in preference to the group, and consequently prefer idiographic approaches to nomothetic ones. The implications of the movement's philosophy for education have been drawn by Carl Rogers (1942, 1945, 1969).

Comparable developments within social psychology may be perceived in the 'science of persons'

movement. It is argued here that we must use ourselves as a key to our understanding of others and, conversely, our understanding of others as a way of finding out about ourselves, an anthropomorphic model of people. Since anthropomorphism means, literally, the attribution of human form and personality, the implied criticism is that social psychology as traditionally conceived has singularly failed, so far, to model people as they really are, and that social science should treat people as capable of monitoring and arranging their own actions, exercising their agency (Harré and Secord, 1972).

Social psychology's task is to understand people in the light of this anthropomorphic model. Proponents of this 'science of persons' approach place great store on the systematic and painstaking analysis of social episodes, i.e. behaviour in context. In Box 1.5 we give an example of such an episode taken from a classroom study. Note how the particular incident would appear on an interaction analysis coding sheet of a researcher employing a positivistic approach. Note, too, how this slice of classroom life can only be understood by knowledge of the specific organizational background and context in which it is embedded.

BOX 1.5 A CLASSROOM EPISODE

Walker and Adelman describe an incident in the following manner:

In one lesson the teacher was listening to the boys read through short essays that they had written for homework on the subject of 'Prisons'. After one boy, Wilson, had finished reading out his rather obviously skimmed piece of work the teacher sighed and said, rather crossly:

T: Wilson, we'll have to put you away if you don't change your ways, and do your homework. Is that all you've done?

P: Strawberries, strawberries. (Laughter)

Now at first glance this is meaningless. An observer coding with Flanders Interaction Analysis Categories (FIAC) would write down:

- '7' (teacher criticizes) followed by a,
- '4' (teacher asks question) followed by a,
- '9' (pupil irritation) and finally a,
- '10' (silence or confusion) to describe the laughter.

Such a string of codings, however reliable and valid, would not help anyone to *understand* why such an interruption was funny. Human curiosity makes us want to know *why* everyone laughs – and so, I would argue, the social scientist needs to know too. Walker and Adelman asked subsequently why 'strawberries' was a stimulus to laughter and were told that the teacher frequently said the pupils' work was 'like strawberries – good as far as it goes, but it doesn't last nearly long enough'. Here a casual comment made in the past has become an integral part of the shared meaning system of the class. It can only be comprehended by seeing the relationship as developing over time.

Source: Adapted from Delamont (1976)

The approach to analysing social episodes in terms of the ‘actors’ themselves is known as the ‘ethogenic method’.⁴ Unlike positivistic social psychology which ignores or presumes its subjects’ interpretations of situations, ethogenic social psychology concentrates on the ways in which persons construe their social world. By probing their accounts of their actions, it endeavours to come up with an understanding of what those persons were doing in the particular episode.

As an alternative to positivist approaches, naturalistic, qualitative, interpretive approaches of various hue possess particular distinguishing features:

- people are deliberate and creative in their actions, they act intentionally and make meanings in and through their activities (Blumer, 1969);
- people actively construct their social world – they are not the ‘cultural dopes’ or passive dolls of positivism (Becker, 1970; Garfinkel, 1967);
- situations are fluid and changing rather than fixed and static; events and behaviour evolve over time and are richly affected by context – they are ‘situated activities’;
- events and individuals are unique and largely non-generalizable;
- a view that the social world should be studied in its natural state, without the intervention of, or manipulation by, the researcher (Hammersley and Atkinson, 1983);
- fidelity to the phenomena being studied is fundamental;
- people interpret events, contexts and situations, and act on the bases of those events (echoing Thomas’s (1928) famous dictum that if people define their situations as real then they are real in their consequences – if I believe there is a mouse under the table, I will act as though there is a mouse under the table, whether there is or not (Morrison, 1998));
- there are multiple interpretations of, and perspectives on, single events and situations;
- reality is multilayered and complex;
- many events are not reducible to simplistic interpretation, hence ‘thick descriptions’ (Geertz, 1973) are essential rather than reductionism; that is to say, thick descriptions representing the complexity of situations are preferable to simplistic ones;
- researchers need to examine situations through the eyes of participants rather than the researcher.

The anti-positivist/post-positivist movement in sociology is represented by three schools of thought – phenomenology, ethnomethodology and symbolic interactionism. A common thread running through the

three schools is a concern with phenomena, that is, the things we directly apprehend through our senses as we go about our daily lives, together with a consequent emphasis on qualitative as opposed to quantitative methodology. The differences between them and the significant roles each phenomenon plays in educational research are such as to warrant a more extended consideration of them in the discussion below.

1.12 A question of terminology: the normative and interpretive paradigms

So far we have introduced and used a variety of terms to describe the numerous branches and schools of thought embraced by the positivist and anti-positivist viewpoints. As a matter of convenience and as an aid to communication, we clarify at this point two generic terms conventionally used to describe these two perspectives and the categories subsumed under each, particularly as they refer to social psychology and sociology. The terms in question are ‘normative’ and ‘interpretive’. The normative paradigm (or model) contains two major orienting ideas (Douglas, 1973): first, that human behaviour is essentially rule-governed; and second, that it should be investigated by the methods of natural science. The interpretive paradigm, in contrast to its normative counterpart, is characterized by a concern for the individual. Whereas normative studies are positivist, theories constructed within the context of the interpretive paradigm tend to be anti-positivist. As we have seen, the central endeavour in the context of the interpretive paradigm is to understand the subjective world of human experience. To retain the integrity of the phenomena being investigated, efforts are made to get inside the person and to understand from within. The imposition of external form and structure is resisted, since this reflects the viewpoint of the observer as opposed to that of the actor directly involved.

Two further differences between the two paradigms may be identified here: the first concerns the concepts of ‘behaviour’ and ‘action’; the second, the different conceptions of ‘theory’. A key concept within the normative paradigm, ‘behaviour’ refers to responses either to external environmental stimuli (e.g. another person, or the demands of society) or to internal stimuli (e.g. hunger, or the need to achieve). In either case, the cause of the behaviour lies in the past. Interpretive approaches, on the other hand, focus on action. This may be thought of as behaviour-with-meaning; it is intentional behaviour, and as such, future oriented. Actions are only meaningful to us insofar as we are able to ascertain the intentions of actors to share their

experiences. A large number of our everyday interactions with one another rely on such shared experiences.

As regards theory (see also Chapter 4), normative researchers try to devise general theories of human behaviour and to validate them through the use of research methodologies which, some believe, push them further and further from the experience and understanding of the everyday world and into a world of abstraction. For them, the basic reality is the collectivity; it is external to the actor and manifest in society, its institutions and its organizations. The role of theory is to say how reality hangs together in these forms or how it might be changed so as to be more effective. The researcher's ultimate aim is to establish a comprehensive 'rational edifice', a universal theory, to account for human and social behaviour.

But what of the interpretive researchers? They begin with individuals and set out to understand their interpretations of the world around them. Indeed they use approaches such as '*verstehen*' ('understanding') and hermeneutic (uncovering and interpreting meanings) to try to see the social world through the eyes of the participants, rather than as an outsider. Here is a view which states that, unlike natural scientists, social scientists recognize that human behaviour is intentional, that people interpret situations through their own eyes and act on those interpretations and that the research has to take cognizance of this. People make sense of the world in their own terms, and such interpretation takes place in socio-cultural, socio-temporal and socio-spatial contexts (cf. Marshall and Rossman, 2016). In turn this requires researchers to suspend or forgo their own assumptions about people, cultures and contexts in favour of looking at a situation and its context in its own terms (cf. Hammersley, 2013, p. 27), to set aside the search for universal statements or causal laws, i.e. to adopt idiographic rather than the nomothetic research of the positivists. The nature of research, then, is exploratory in nature, to investigate the interpretations of the situation made by the participants themselves, to understand their attitudes, behaviours and interactions.

In interpretive research, theory is emergent and arises from particular situations; it is 'grounded' in data generated by the research (Glaser and Strauss, 1967) (see Chapter 37). Theory should not precede research but follow it. Investigators work directly with experience and understanding to build their theory on them. The data thus yielded will include the meanings and purposes of those people who are their source. Further, the theory so generated must make sense to those to whom it applies. The aim of scientific investigation for the interpretive researcher is to understand how this

reality goes on at one time and in one place and compare it with what goes on in different times and places. Thus theory becomes sets of meanings which yield insight and understanding of people's behaviour. These theories are likely to be as diverse as the meanings and understandings that they seek to explain. From an interpretive perspective, the hope of a universal theory which characterizes the normative outlook gives way to multifaceted images of human behaviour as varied as the situations and contexts supporting them.

1.13 Phenomenology, ethnomethodology, symbolic interactionism and constructionism

There are many variants of qualitative, naturalistic, interpretive approaches (Hitchcock and Hughes, 1995). Marshall and Rossman (2016) identify several such 'genres' (pp. 17–41). Under 'major genres' they include those which: (a) focus on culture and society (e.g. ethnographic approaches); (b) focus on the lived experiences of individuals (phenomenological approaches); (c) focus on texts and talking (sociolinguistic approaches); (d) use grounded theory approaches; and (e) use case studies. Under 'critical genres' they include: (a) critical ethnography and autoethnography; (b) critical discourse analysis; (c) action research and participatory action research; (d) queer theory; (e) critical race theory; (f) feminist theory; (g) cultural studies; and (h) internet/virtual ethnography. We discuss critical theories in Chapter 3. Here we focus on four significant 'traditions' in the interpretive style of research – phenomenology, ethnomethodology, symbolic interactionism and constructionism.

Phenomenology

In its broadest meaning, phenomenology is a theoretical point of view that advocates the study of direct experience taken at face value and which sees behaviour as determined by the phenomena of experience rather than by external, objective and physically described reality (English and English, 1958). Although phenomenologists differ among themselves on particular issues, there is fairly general agreement on the following points identified by Curtis (1978), Hammersley (2013) and Marshall and Rossman (2016), which can be taken as distinguishing features of their philosophical viewpoint:

- A belief in the importance, and even the primacy, of subjective consciousness;
- The importance of documenting and describing immediate experiences;

- The significance of understanding how and why participants' knowledge of a situation comes to be what it is;
- The social and cultural situatedness of actions and interactions, together with participants' interpretations of a situation;
- An understanding of consciousness as active, as meaning bestowing;
- A claim that there are certain essential structures to consciousness of which we gain direct knowledge by a certain kind of reflection. Exactly what these structures are is a point about which phenomenologists differ.

Various strands of development may be traced in the phenomenological movement: we briefly examine two of them – the transcendental phenomenology of Husserl; and existential phenomenology, of which Schutz is perhaps the most characteristic representative.

Husserl, regarded by many as the founder of phenomenology, was concerned with investigating the source of the foundation of science and with questioning the common-sense, 'taken-for-granted' assumptions of everyday life (see Burrell and Morgan, 1979). To do this, he set about opening up a new direction in the analysis of consciousness. His catchphrase was 'back to the things!' which for him meant finding out how things appear directly to us rather than through the media of cultural and symbolic structures. In other words, we are asked to look beyond the details of everyday life to the essences underlying them. To do this, Husserl exhorts us to 'put the world in brackets' or free ourselves from our usual ways of perceiving the world. What is left over from this reduction is our consciousness, of which there are three elements – the 'I' who thinks, the mental acts of this thinking subject, and the intentional objects of these mental acts. His was a call to overcome the subjective–objective divide. The aim, then, of this method of *epoché*, as Husserl called it, is the dismembering of the constitution of objects in such a way as to free us from all preconceptions about the world.

Schutz was concerned with relating Husserl's ideas to the issues of sociology and to the scientific study of social behaviour. Of central concern to him was the problem of understanding the meaning structure of the world of everyday life. He sought the origins of meaning in the 'stream of consciousness' – basically an unbroken stream of lived experiences which have no meaning in themselves. One can only impute meaning to them retrospectively, by the process of turning back on oneself and looking at what has been going on. In other words, meaning can be accounted for here by the

concept of reflexivity. For Schutz, the attribution of meaning reflexively is dependent on the people identifying the purpose or goal they seek (Burrell and Morgan, 1979).

According to Schutz, the way we understand the behaviour of others is dependent on a process of typification by means of which the observer makes use of concepts resembling 'ideal types' to make sense of what people do. These concepts are derived from our experience of everyday life and it is through them, claims Schutz, that we classify and organize our everyday world. In this respect he adhered to principles of empiricism. As Burrell and Morgan observe, we learn these typifications through our biographical locations and social contexts. Our knowledge of the everyday world inheres in social order and itself is socially ordered.

The fund of everyday knowledge by means of which we are able to typify other people's behaviour and come to terms with social reality varies from situation to situation. We thus live in a world of multiple realities, and social actors move within and between these, abiding by the rules of the game for each of these worlds.

Ethnomethodology

Like phenomenology, ethnomethodology is concerned with the world of everyday life, studying participants' circumstances, thoughts and commonplace daily lives as worthy of empirical study (Garfinkel, 1967, p. vii). Garfinkel maintains that students of the social world must doubt the reality of that world; and that in failing to view human behaviour more sceptically, sociologists have created an ordered social reality that bears little relationship to the real thing. He thereby challenges the basic sociological concept of order.

Ethnomethodology, then, is concerned with how people make sense of their everyday world. More especially, it is directed at the mechanisms by which participants achieve and sustain interaction in a social encounter – the assumptions they make, the conventions they utilize, and the practices they adopt. Ethnomethodology thus seeks to understand social accomplishments in their own terms; it is concerned to understand them from within (Burrell and Morgan, 1979).

In identifying the 'taken-for-granted' assumptions characterizing a social situation and the ways in which the people involved make their activities rationally accountable, ethnomethodologists use notions of 'indexicality' and 'reflexivity'. Indexicality refers to the ways in which actions and statements are related to the social contexts producing them, and to the way

their meanings are shared by the participants but not necessarily stated explicitly. Indexical expressions are thus the designations imputed to a particular social occasion by the participants in order to locate the event in the sphere of reality. Reflexivity, on the other hand, refers to the way in which all accounts of social settings – descriptions, analyses, criticisms etc. – and the social settings occasioning them, are mutually inter-dependent.

One can distinguish between two types of ethnomethodologists: *linguistic* and *situational*. *Linguistic* ethnomethodologists focus upon the use of language and the ways in which conversations in everyday life are structured. Their analyses make much use of the unstated ‘taken-for-granted’ meanings, the use of indexical expressions and the way in which conversations convey much more than is actually said. *Situational* ethnomethodologists cast their view over a wider range of social activity and seek to understand the ways in which people negotiate the social contexts in which they find themselves. They are concerned to understand how people make sense of and order their environment. As part of their empirical method, ethnomethodologists may consciously and deliberately disrupt or question the ordered ‘taken-for-granted’ elements in everyday situations in order to reveal the underlying processes at work.

The substance of ethnomethodology thus largely comprises a set of specific techniques and approaches to be used in studying what Garfinkel has described as the ‘awesome indexicality’ of everyday life. It is geared to empirical study, and the stress which its practitioners place upon the uniqueness of the situation encountered projects its essentially relativist standpoint. A commitment to the development of methodology and fieldwork has occupied first place in the interests of its adherents, so that related issues of ontology, epistemology and the nature of human beings have received less attention than perhaps they deserve.

Symbolic interactionism

Essentially, the notion of symbolic interactionism derives from the work of Mead (1934). Although subsequently to be associated with such noted researchers as Blumer, Hughes, Becker and Goffman, the term does not represent a unified perspective in that it does not embrace a common set of assumptions and concepts accepted by all who subscribe to the approach. Here, however, it is possible to identify three basic postulates. These have been set out by Woods (1979) as follows. First, human beings act towards things on the basis of the meanings they have for them. Humans inhabit two different worlds: the ‘natural’ world wherein they are

organisms of drives and instincts and where the external world exists independently of them, and the social world where the existence of symbols, like language, enables them to give meaning to objects. This attribution of meanings, this interpreting, is what makes them distinctively human and social. Interactionists therefore focus on the world of subjective meanings and the symbols by which they are produced and represented. This means not making any prior assumptions about what is going on in an institution, and taking seriously, indeed giving priority to, inmates’ own accounts. Thus, if students appear preoccupied for too much of the time – ‘being bored’, ‘having a laugh’ etc. – the interactionist is keen to explore the properties and dimensions of these processes.

Second, this attribution of meaning to objects through symbols is a continuous process. Action is not simply a consequence of psychological attributes such as drives, attitudes or personalities, or determined by external social facts such as social structure or roles, but results from a continuous process of meaning attribution which is always emerging, in a state of flux and subject to change. The individual constructs, modifies, pieces together, weighs up the pros and cons, and bargains.

Third, this process takes place in a social context. Individuals align their actions to those of others. They do this by ‘taking the role of the other’, by making indications to themselves about others’ likely responses. They construct how others wish to or might act in certain circumstances, and how they themselves might act. They might try to ‘manage’ the impressions others have of them, put on a ‘performance’, try to influence others’ ‘definition of the situation’.

Instead of focusing on the individual, then, and his or her personality characteristics, or on how the social structure or social situation causes individual behaviour, symbolic interactionists direct their attention at the nature of interaction, the dynamic activities taking place between people. In focusing on the interaction itself as a unit of study, the symbolic interactionist creates a more active image of the human being and rejects the image of the passive, determined organism. Individuals interact; societies are made up of interacting individuals. People are constantly undergoing change in interaction and society is changing through interaction. Interaction implies human beings acting in relation to each other, taking each other into account, acting, perceiving, interpreting, acting again. Hence, a more dynamic and active human being emerges rather than an actor merely responding to others. Woods (1983, pp. 15–16) summarizes key emphases of symbolic interaction thus:

- individuals as constructors of their own actions;
- the various components of the self and how they interact; the indications made to self, meanings attributed, interpretive mechanisms, definitions of the situation; in short, the world of subjective meanings, and the symbols by which they are produced and represented;
- the process of negotiation, by which meanings are continually being constructed;
- the social context in which they occur and whence they derive;
- by taking the ‘role of the other’ – a dynamic concept involving the construction of how others wish to or might act in a certain circumstance, and how individuals themselves might act – individuals align their actions to those of others.

Constructionism

In constructionism (also termed constructivism), in contrast to the argument that external objects and factors determine, shape, impress, print or fix themselves onto passive recipients (i.e. are ‘givens’ in society or individuals), people actively and agentically seek out, select and construct their own views, worlds and learning, and these processes are rooted in socio-cultural contexts and interactions. In other words, cognition is generative and active rather than receptive and passive respectively. Through such active cognition and deliberate perception we come to understand ourselves and how this affects the worlds we inhabit and the way in which we interact with the objects and people in them.

Hammersley (2013) notes that constructionism requires researchers to focus on the processes that lead to the construction, constitution and character given to independent objects and the relationships between them (pp. 35–6), i.e. how people collectively construct their social worlds (e.g. through discourse analysis) (p. 36). He gives an example of replacing the definition of a person as ‘intelligent’ with an examination of the ‘discursive practices’ which led to the construction of that person being intelligent and how this affects how that person operates in socio-cultural and institutional contexts (p. 36).

Social constructionism holds that individuals seek to make meaning of their social lives and that the researcher has to examine the situation in question through the multiple lenses of the individuals involved, to obtain their definition of the situation, to see how they make sense of their situation and to focus on interactions, contexts, environments and biographies. Indeed social constructionism emphasizes the social nature of learning, arguing that it is only through social

interaction and communication that certain types of learning occur and certain views of the world are constructed.

A characteristic common to the phenomenological, ethnomethodological, symbolic interactionist and constructionist perspectives, which makes them attractive to the educational researcher, is the way they fit naturally to the kind of concentrated action found in classrooms and schools. Yet another shared characteristic is the manner in which they are able to preserve the integrity of the situation in which they are employed. Here the influence of the researcher in structuring, analysing and interpreting the situation is present to a much smaller degree than would be the case with a more traditionally oriented research approach.

1.14 Criticisms of the naturalistic and interpretive approaches

Critics have wasted little time in pointing out what they regard as weaknesses in these newer qualitative perspectives. They argue that while it is undeniable that our understanding of the actions of our fellow-beings necessarily requires knowledge of their intentions, this, surely, cannot be said to constitute *the* purpose of a social science. As Rex observed:

Whilst patterns of social reactions and institutions may be the product of the actors’ definitions of the situations there is also the possibility that those actors might be falsely conscious and that sociologists have an obligation to seek an objective perspective which is not necessarily that of any of the participating actors at all.... We need not be confined purely and simply to that ... social reality which is made available to us by participant actors themselves.

(Rex, 1974)

While these more recent perspectives have presented models of people that are more in keeping with common experience, some argue that anti-positivists/post-positivists have gone too far in abandoning scientific procedures of verification and in giving up hope of discovering useful generalizations about behaviour. Are there not dangers in rejecting the approach of physics in favour of methods more akin to literature, biography and journalism? Some specific criticisms of the methodologies are well directed, for example Argyle (1978) questions whether, if carefully controlled interviews such as those used in social surveys are inaccurate, then the less controlled interviews carry even greater risks of inaccuracy. Indeed Bernstein (1974) suggests that subjective reports may be incomplete and misleading.

I may believe that the teacher does not like me, and, therefore, act as though the teacher does not like me (a self-fulfilling prophecy), but, in fact, all the time the teacher actually does like me; my perception is wrong.

Bernstein's criticism is directed at the overriding concern of phenomenologists and ethnomethodologists with the meanings of situations and the ways in which these meanings are negotiated by the actors involved. What is overlooked about such negotiated meanings, observes Bernstein, is that the very process whereby one interprets and defines a situation is itself a product of the circumstances in which one is placed. One important factor in such circumstances that must be considered is the power of others to impose their own definitions of situations upon participants. Doctors' consulting rooms and headteachers' studies are locations in which inequalities in power are regularly imposed upon unequal participants. The ability of certain individuals, groups, classes and authorities to persuade others to accept their definitions of situations demonstrates that while – as ethnomethodologists insist – social structure is a consequence of the ways in which we perceive social relations, it is clearly more than this.

Conceiving of social structure as external to ourselves helps us include its self-evident effects upon our daily lives into our understanding of the social behaviour going on about us. Here is rehearsed the tension between agency and structure of social theorists (Layder, 1994); the danger of interactionist and interpretive approaches is their relative neglect of the power of external – structural – forces to shape behaviour and events. There is a risk in interpretive approaches that they become hermetically sealed from the world outside the participants' theatre of activity – they put artificial boundaries around subjects' behaviour. Just as positivistic theories can be criticized for their macro-sociological persuasion, so interpretive and qualitative theories can be criticized for their narrowly micro-sociological perspectives.

1.15 Postmodernist and post-structuralist perspectives

It is not only post-positivists who challenge the modernist, positivist conception of the world. For modernists the world is available to be studied objectively and, by using scientific methods, to arrive at secure, rigorous, scientific, discipline-based explanations of observed phenomena – 'grand narratives' which are redolent of the Enlightenment project of providing foundationalist and absolute knowledge. Postmodernism challenges each of these. Whilst it is perhaps invidious to try to characterize postmodernists (as they

would argue against any singular or all-embracing definitions), in a seminal text Jameson (1991) argues that postmodernism does have several distinguishing hallmarks, including, for example:

- the absence of 'grand narratives' (metanarratives) and grand designs, laws and patterns of behaviour (thereby, ironically, eclipsing the status of their own narrative);
- the valorization of discontinuity, difference, diversity, pluralism, variety, uniqueness, subjectivity, distinctiveness and individuality;
- the importance of the local, the individual and the particular;
- the 'utter forgetfulness of the past' and the 'autoreferentiality' of the present (Jameson, 1991, p. 42);
- the importance of temporality and context in understanding phenomena: meanings are rooted in time, space, cultures, societies and are not universal across these;
- the celebration of depthlessness, multiple realities (and, as Jameson argues, multiple superficialities) and the rectitude of individual interpretations and meanings rather than an appeal to a singular or universal rationalism;
- relativism rather than absolutism in deciding what constitutes worthwhile knowledge, research and their findings;
- the view of knowledge as a human, social construct;
- multiple, sometimes contradictory, yet co-existent interpretations of the world, in which the researcher's interpretation is only one out of several possible interpretations, i.e. the equal value of different interpretations and the reduction in the authority of the researcher, yet, simultaneously, the privileging of some interpretations of the world to the neglect of others (i.e. the nexus between knowledge and power, a feature of critical theory, discussed in Chapter 3);
- the recognition that researchers are part of the world that they are researching;
- the emancipatory potential of according value to individual views, values, perspectives and interpretations (see Chapter 3).

Pring (2015) adds to this the point that postmodernism is characterized by a revolt against thought control and cultural control, by an assertion of multiple forms of cultural expression, an abandonment of certainty, a replacement of 'authority' (as in 'authoritative') by multiple voices and negotiated meanings, and a blurring of artificial boundaries (disciplines) of knowledge, a questioning of received wisdoms and a recognition of

fallibilism, all of which he sees as the function of the ‘perennial philosophical tradition’ and not one given birth to by postmodernism (pp. 134–7). In one sense postmodernism supports the interpretive paradigm set out earlier in this chapter. In another sense it supports complexity theory as discussed below, and in a third sense it supports critical theory as set out in Chapter 3. Postmodernism has a chameleon-like nature in this respect.

Post-structuralism, like postmodernism, has many different interpretations (we will not discuss here the interpretation that relates to semiology). Here we take a necessarily selective interpretation, to focus on those features that are relevant to the foundations and conduct of educational research. Here post-structuralism can be regarded as a counter to those structural-functionalists who adopt a systems view of society (e.g. Marxism, or functionalist anthropologists such as Lévi-Strauss) or behaviour as a set of interrelated parts which, in law-like fashion, pattern themselves and fit together neatly into a fixed view of the world and its operations and in which individual behaviour is largely determined by given, structural features of society (e.g. social class, position in society, role in society). In post-structuralist approaches, data (e.g. conversations, observations) and even artefacts can be regarded as texts (Burman and Parker, 1993), as discourses that are constructed and performed through discourses (see Chapter 35), open to different meaning and interpretations (Francis, 2010, p. 327).

Post-structuralists (e.g. Foucault, Derrida) argue that individual agency has prominence; individuals are not simply puppets of a given system; people are diverse and different, indeed they may carry contradictions and tensions within themselves (e.g. in terms of class, ethnicity, gender, employment, social group, family membership and tasks, and so on); they are not simply the decentred bearers of given roles. Individuals have views of themselves, and one task of the researcher is to locate research findings within the views of the self that the participants hold, and to identify the meanings which the participants accord to phenomena. Hence not only do the multiple perspectives of the participants have to be discerned, but also those of the researchers, the audiences of the research and the readers of research. The task of the research is to ‘deconstruct’, to expose, the different meanings, layers of meanings and privileging of meanings inherent in a phenomenon or piece of research. There is no single, ‘essential’ meaning, but many, and one task of research is to understand how meanings and knowledge are produced, legitimized and used. (This links post-structuralism to critical theory, though some critical theorists, e.g. Habermas (1987), argue against

critical theory’s affinity to postmodernism or post-structuralism.)

One can detect affinities between post-positivism, postmodernism and post-structuralism in underpinning interpretive and qualitative approaches to educational research, complexity theory and critical theory, and the significance given to individual and subjective accounts in the research process, along with reflexivity on the part of the researcher. (That said, many post-positivists, postmodernists and post-structuralists would reject such a simple affinity, or even the links between their views and, for example, phenomenology and interpretivism. We do not explore this here.) One can suggest that post-positivism, postmodernism and post-structuralism argue for multiple interpretations of a phenomenon to be provided, to accord legitimacy to individual voices in research, and to abandon the search for deterministic, simple cause-and-effect laws of behaviour and action.

1.16 Subjectivity and objectivity in educational research

The preceding overview has alluded to the sympathies between some paradigms and objectivity in research and other paradigms and subjectivity in research. To make such an exclusive separation is a chimera, a false dichotomy. With regard to objectivity, to say, for instance, that objectivity inheres in positivist and post-positivist approaches overlooks not only the several interpretations of positivism and post-positivism but what it means to be subjective. Objectivity is refracted through the researcher’s eyes and the generation, construction and testing of hypotheses draw on personal understandings and formulations. In other words, objectivity cannot escape some subjective roots. Taken to an extreme, it leads to a rejection of the idea that the researcher can ever be objective, just as there is a rejection of the idea that there is an objective reality or ‘truth’ about a phenomenon (Hammersley, 2011, p. 89). Objectivity here is defined as intersubjectivity (as opposed to subjectivity), reliability and freedom from bias (Risjord, 2014, p. 22). Risjord illustrates the difference between intersubjectivity and subjectivity thus (p. 23): I feel hungry (subjective) so I eat a sandwich (intersubjective, in that it can be seen by an observer, i.e. is open to critical scrutiny).

On the other hand, subjectivity cannot turn its back on what is ‘out there’ in terms of overriding the social, societal and institutional social facts, which have an existence independent of the participant. Subjectivity cannot lay claim to being a privileged discourse without risking relativism.

Subjectivity and objectivity are frequently placed at the poles of different continua (cf. Hammersley, 2011, p. 90), for example:

<i>Subjective</i>		<i>Objective</i>
Internal	...	External
Private	...	Public
Positivist	...	Interpretive
Idiographic	...	Nomothetic
Judgement	...	Technical application (e.g. of statistics)
Individual	...	Shared
Personal	...	Impersonal
Particular	...	General
Relative	...	Absolute
Opinion	...	Proof
Experimental	...	Interactionist
Biased	...	Bias-free
Unobservable	...	Observable
Idiosyncratic	...	Regular
Uncertain	...	Certain
Unpredictable	...	Predictable
Unreliable	...	Reliable
Imprecise	...	Explicit
Questionable	...	Conclusive
Unverifiable	...	Checkable
Prone to error	...	Secure
Complex	...	Straightforward
Opaque	...	Transparent

Source: Adapted from Barr Greenfield (1975)

However, this creates false dichotomies, and look how easily one can create biases in the pejorative terms used: many of the items in the left-hand column are presented as the shabby, less respectable end of research, whilst the right-hand column seems much more clean and respectable. This can overlook the risk of bias and errors that researchers might commit in working in the right-hand column and the authenticity, correctness and truth of the left-hand column. Both subjective and objective views have to face judgements of plausibility, validity, reliability, meaningfulness and credibility.

However, more fundamentally, as Hammersley (2011) remarks, we depend on personal knowledge and judgement in making meaning of phenomena and data, be those data numbers, words, pictures or sounds. We rely on our senses in making observations. Following objective procedures requires a personal commitment.

We rely on our judgement in raising hypotheses, making inferences and drawing conclusions. However, simply amassing subjective data from participants does not ensure that the data are true or reliable, but stating objective procedures does not ensure identical practices, not least as, in the social world, researchers – consciously or not – adjust their practices to the situation and the people who are participating in the research; standardizing practice has to extend to participants.

Medical research is a good example here: whilst there might be an objective, standardized procedure for patients taking medicine in a randomized controlled trial, that does not guarantee that patients will follow it: they might refuse to take the medicine, forget to take it, take it at the wrong time of day, take some but not all of it, take the wrong dose (too little or too much), misread the instructions, and so on. Intention does not match actuality.

The claims we make from knowledge, be they from the left-hand or right-hand columns here, do not constitute absolute truth: the same data can, and do, sustain multiple interpretations, claims and conclusions. Further, is it really possible or desirable to set aside one's own biography, values and assumptions, however reflexive one might be? Reflexivity is not the same as objectivity. Is it not the case, anyway, that knowledge, particularly of the social world, is a socio-temporal construction rather than the clean world of the objectivist, and to pretend otherwise is simply naive or deceitful (Hammersley, 2011, p. 96)? Or is this giving in to the relativists and the postmodernists, in the knowledge that relativism is, by its own definition, only relative, and that the postmodernists cannot lay claim to their views as having any status at all as to do so would be to acknowledge that metanarratives exist – a claim which postmodernists proscribe as an article of faith.

Hammersley (2011) is clear that errors may stem from the researcher's own social or individual characteristics and their influence on their research, but that it is unnecessary and, indeed, undesirable to assume that the researcher can stand out completely from his or her social and individual characteristics. Further, error does not automatically follow from an acknowledgement of the researcher's own social and individual characteristics.

The task, then, is to protect the research from negative effects of subjectivity (2011, p. 101), though Hammersley acknowledges that what constitutes 'error' is not always clear. However, he offers researchers some advice here, cautioning them to be on their guard against preconceptions, prior assumptions, preferences and biases that are 'external to the pursuit of knowledge' (p. 102), i.e. which are goals that are separate

from the research itself. Objectivity, in this case, means adhering to the ‘epistemic virtue’ of keeping *only* to the canons and requirements of the research itself, setting aside any extraneous personal convictions or subordinating the research to any other goals outside the research (p. 103). Given this, objectivity and the suppression of personal, subjective beliefs, values, commitments or agendas have a key role to play in educational research. The objective reliability of the research does not depend on the political, evaluative or moral motivations of the researcher (cf. Risjord, 2014, p. 23).

Similarly, value-neutrality in educational and social science research leaves unsaid any comment on what *ought* or *ought not* be done; that is for policy makers. Rather, educational research confines itself to facts; that is, the scientific enterprise. Saying that teachers should not assault students is an evaluative statement and not a matter for social science research, as it does not rest on empirical data alone, though reporting incidents of assault and its effects surely is a matter for research.

Whether researchers should have a ‘committed’ position is a matter that we return to in Chapter 3 on critical theory, which explicitly disavows value-free positions, and argues for partisan positions in research as contributing to the greater good of an emancipated society in freeing itself from that ideology which conceals oppression and unjust subordination and power differentials of social groups, and which transforms society to equality, democracy and social justice. Fact and value reunite.

1.17 The paradigm of complexity theory

An emerging paradigm in educational research is that of complexity theory (Medd, 2002; Morrison, 2002a, 2008; Radford, 2006, 2007, 2008; Kuhn, 2007; Byrne and Callaghan, 2014; Boulton *et al.*, 2015), as schools can be regarded as ‘complex adaptive systems’ (Kaufmann, 1995). Complexity theory looks at the world in ways which break with simple cause-and-effect models, simple determinism and linear predictability (Morrison, 2008) and a dissection/atomistic approach to understanding phenomena (Radford, 2007, 2008; Byrne and Callaghan, 2014), replacing them with organic, non-linear and holistic approaches (Santonus, 1998, p. 3). Relations within interconnected, dynamic and changing networks are the order of the day (Wheatley, 1999, p. 10), and there is a ‘multiplicity of simultaneously interacting variables’ (Radford, 2008, p. 510). Here key terms are feedback, recursion, emergence, connectedness and self-organization. Out go the

simplistic views of linear causality (Radford, 2007; Morrison, 2009; Byrne and Callaghan, 2014; Boulton *et al.*, 2015), the ability to predict, control and manipulate, to apply reductive techniques to research, and in come uncertainty, networks and connection, holism self-organization, emergence over time through feedback and the relationships of the internal and external environments, and survival and development through adaptation and change.

In complexity theory, a self-organizing system is autocatalytic and possesses its own unique characteristics and identity (Kelly and Allison, 1999, p. 28) which enable it to perpetuate and renew itself over time – it creates the conditions for its own survival. This takes place through engagement with others in a system (Byrne and Callaghan, 2014; Boulton *et al.*, 2015). The system is aware of its own identity and core properties, and is self-regenerating (able to sustain that identity even though aspects of the system may change, e.g. staff turnover in a school).

Through feedback, recursion, perturbation, autocatalysis, connectedness and self-organization, higher levels of complexity and differentiated, new forms of life, behaviour and systems arise from lower levels of complexity and existing forms. These complex forms derive from often comparatively simple sets of rules – local rules and behaviours generating emergent complex global order and diversity (Waldrop, 1992, pp. 16–17; Lewin, 1993, p. 38). General laws of emergent order can govern adaptive, dynamical processes (Waldrop, 1992, p. 86; Kauffman, 1995, p. 27).

The interaction of individuals feeds into the wider environment which, in turn, influences the individual units of the network; they co-evolve, shaping each other (Stewart, 2001), and co-evolution requires connection, cooperation and competition: competition to force development and cooperation for mutual survival. The behaviour of a complex system as a whole, formed from its several elements, is greater than the sum of the parts (Byrne and Callaghan, 2014; Boulton *et al.*, 2015).

Feedback occurs between the interacting elements of the system. Negative feedback is regulatory (Marion, 1999, p. 75), for example learning that one has failed a test. Positive feedback brings increasing returns and uses information to change, grow and develop (Wheatley, 1999, p. 78); it amplifies small changes (Stacey, 1992, p. 53). Once a child has begun to read she is gripped by reading, she reads more and learns at an exponential rate.

Connectedness, a key feature of complexity theory, exists everywhere. In a rainforest ants eat leaves, birds eat ants and leave droppings, which fertilize the soil for

growing trees and leaves for the ants (Lewin, 1993, p. 86). In schools, children are linked to families, teachers, peers, societies and groups; teachers are linked to other teachers, support agencies (e.g. psychological and social services), policy-making bodies, funding bodies, the legislature, and so on. The child (indeed the school) is not an island, but is connected externally and internally in several ways. Disturb one element and the species or system must adapt or die; the message is ruthless.

Emergence is the partner of *self-organization*. Systems possess the ability for self-organization, which is not according to an a priori grand design – a cosmological argument – nor a teleological argument; complexity is neither. Further, self-organization emerges, it is internally generated; it is the opposite of external control. As Kauffman (1995) suggests, order comes for free and replaces control. Order is not imposed; it emerges; in this way it differs from control. Self-organized order emerges of itself as the result of the interaction between the organism and its environment, and new structures emerge that could not have been predicted; that emerged system is, itself, complex and cannot be reduced to those parts that gave rise to the system. As Davis and Sumara (2005, p. 313) write: ‘phenomena have to be studied at their level of emergence’, i.e. at their present overall state, not in terms of the elements present in the pre-metamorphosed state.

Stacey (2000) suggests that a system can only evolve, and evolve spontaneously, where there is diversity and deviance (p. 399) – a salutary message for command-and-control teachers who exact compliance from their pupils. The future is largely unpredictable. At the point of ‘self-organized criticality’ (Bak, 1996), a tipping point, the effects of a single event are likely to be very large, breaking the linearity of Newtonian reasoning wherein small causes produce small effects; the straw that breaks the camel’s back.

Complexity theories argue against the linear, deterministic, patterned, universalizable, stable, atomized, modernistic, objective, mechanist, controlled, closed systems of law-like behaviour which may be operating in the laboratory but which do not operate in the social world of education. These features of complexity theories seriously undermine the value of experiments and positivist research in education (e.g. Waldrop, 1992; Lewin, 1993).

Complexity theory replaces these with an emphasis on networks, linkages, holism, feedback, relationships and interactivity in context (Byrne and Callaghan, 2014), emergence, dynamical systems, self-organization and an open system (rather than the closed world of the experimental laboratory). Even if one could conduct an

experiment, its applicability to ongoing, emerging, interactive, relational, open situations, in practice, is limited (Morrison, 2001). It is misconceived to hold variables constant in a dynamical, evolving, fluid, open situation. What is measured is history.

Complexity theory challenges randomized controlled trials – the ‘gold standard’ of research. Classical experimental methods, abiding by the need for replicability and predictability, may not be particularly fruitful since, in complex phenomena, results are never clearly replicable or predictable: As Heraclitus noted, we never jump into the same river twice. Complexity theory suggests that educational research should concern itself with: (a) how multivalency and non-linearity feature in education; (b) how voluntarism and determinism, intentionality, agency and structure, life-world and system, divergence and convergence interact in learning (Morrison, 2002a, 2005); (c) how to both use, but transcend, simple causality in understanding the processes of education (Morrison, 2012); (d) how viewing a system holistically, as having its own ecology of multiple interacting elements, is more powerful than an atomized approach. Complexity theory suggests that phenomena must be looked at holistically; to atomize phenomena into measurable variables and then to focus only on certain of these is to miss synergy, the dynamic interaction of several parts (Morrison, 2008) and the significance of the whole. Measurement, however acute, may tell us little of value about a phenomenon; one can measure every observable variable of a person to an infinitesimal degree, but his/her nature, what makes him/her who he or she is, eludes atomization and measurement.

These should merge, so that in complexity theory the unit of analysis becomes a web, network or ecosystem (Capra, 1996, p. 301; Morrison, 2012), focused on, and arising from, a specific topic or centre of interest (a ‘strange attractor’). Individuals, families, students, classes, schools, communities and societies exist in symbiosis; complexity theory tells us that their relationships are necessary, not contingent, and analytic, not synthetic. This is a challenging prospect for educational research, and complexity offers considerable leverage into understanding societal, community, individual and institutional change theory (Radford, 2006; Morrison, 2008); it provides the nexus between macro- and micro-research in understanding and promoting change.

In addressing holism, complexity theory suggests the need for case study methodology, narrative approaches, action research and participatory forms of research, premised in many ways on interactionist, qualitative accounts, i.e. looking at situations through the eyes of as many participants or stakeholders as

possible (e.g. Byrne and Callaghan, 2014; Boulton *et al.*, 2015). This enables multiple causality, multiple perspectives and multiple effects to be charted (Morrison, 2012). Self-organization, a key feature of complexity theory, argues for participatory, collaborative and multi-perspectival approaches to educational research. This is not to deny ‘outsider’ research; it is to suggest that, if it is conducted, outsider research has to take in as many perspectives as possible.

In educational research terms, complexity theory stands against methodologies based on linear views of causality, arguing for multiple causality, multi-directional causes and effects and networks of causes (Morrison, 2012) at a host of different levels and in a range of diverse ways. No longer can one be certain that a simple cause brings a simple or single effect, or that a single effect is the result of a single cause, or that the location of causes will be in single fields only, or that the location of effects will be in a limited number of fields (Morrison, 2009, 2012). Researching causality becomes a search for networked, multi-causality and multi-stranded causality (Morrison, 2012).

Complexity theory not only questions the values of positivist research and experimentation, but it also underlines the importance of educational research to catch the deliberate, intentional, agentic actions of participants and to adopt interactionist and constructivist perspectives. (In this respect it has sympathies, perhaps, with posthumanism, though it is a very different animal from posthumanism.) Kuhn (2007, pp. 172–3) sets out a series of axioms for complexity-based research: (a) reality is dynamic, emergent and self-organizing, requiring multiple perspectives to be addressed (see also Medd, 2002); (b) the relationship between the knower and the known is, itself, dynamic, emergent and self-organizing; (c) hypotheses for research must relate to time and context (cf. Medd, 2002; Radford, 2006); (d) it is impossible to distinguish cause from effect, as entities are mutually shaping and influencing (co-evolution); (e) inquiry is not value-free.

Addressing complexity theory’s argument for self-organization, the call is for the teacher-as-researcher movement to be celebrated, and complexity theory suggests that research in education could concern itself with the symbiosis of internal and external researchers and research partnerships. Just as complexity theory suggests that there are multiple views of reality, so this accords not only with the need to catch several perspectives on a situation (using multi-methods), but resonates with those tenets of critical research which argue for different voices, views and interpretations to be heard, incorporated and understood respectively. Heterogeneity is the watchword.

Complexity theory provides not only a powerful challenge to conventional approaches to educational research, but it suggests both a substantive agenda and also a set of methodologies, arguing for methodological, paradigmatic and theoretical pluralism. For example, Byrne and Callaghan (2014) and Boulton *et al.* (2015) suggest that research should study the processes of emergence over time and critical incidents in evolving situations. In addressing holism, complexity theory suggests the need for case study methodology, qualitative research and participatory, multi-perspectival and collaborative (self-organized), partnership-based forms of research, premised on interactionist, qualitative and interpretive accounts (e.g. Lewin and Regine, 2000).

1.18 Conclusion

This chapter has argued that planning and conducting educational research cannot follow simple recipes but is a complex, deliberative and iterative process in which ontological and epistemological matters have to be considered and in which many different kinds of understanding feature. In addressing this, the chapter has introduced several paradigms and their possible contribution to educational research, including: positivism, post-positivism, post-structuralism, postmodernism and complexity theory. It has commented on different views of social reality and a range of approaches to understanding that reality: deductive and inductive; empirical and rationalist; nomothetic and idiographic; subjective and objective; the scientific method; and alternatives in naturalistic, interpretive, phenomenological, interactionist and constructionist approaches.

The argument through the chapter has suggested that foundationalism and the quest for absolute knowledge in educational research is questionable. In this it has indicated the expanding range of approaches, of which, for example, postmodernism, post-structuralism and complexity theory are examples. Complexity theory challenges conceptions of simple cause-and-effect, experimental approaches to research and it advocates attention to context and holism in educational research.

In recognizing the many and expanding number of paradigms and approaches to educational research, the chapter has argued for methodological, paradigmatic and theoretical pluralism, indeed mixed methods (Chapter 2). These set the ground for the many approaches, designs, methodologies and methods set out in the remainder of the book. Simple recipe-following is out, and deliberation, fitness for purpose and fitness of purpose are key watchwords here.

The companion website to the book provides additional material and PowerPoint slides for this chapter, which list the structure of the chapter and then provide a summary of the key points in each of its sections. This resource can be found online at: www.routledge.com/cw/cohen.

Notes

- 1 We are not here recommending, nor would we wish to encourage, exclusive dependence on rationally derived and scientifically provable knowledge for the conduct of education – even if this were possible. There is a rich fund of traditional and cultural wisdom in teaching (as in other spheres of life) which we would ignore to our detriment. What we are suggesting, however, is that total dependence on the latter has tended in the past to lead to an impasse, and that for further development and greater understanding to be achieved education must needs resort to the methods of science and research.
- 2 A classic statement opposing this particular view of science is that of Kuhn (1962), *The Structure of Scientific Revolutions*. Kuhn's book, acknowledged as an intellectual tour de force, makes the point that science is not the systematic accumulation of knowledge as presented in textbooks; that it is a far less rational exercise than generally imagined. In effect, 'it is a series of peaceful interludes punctuated by intellectually violent revolutions ... in each of which one conceptual world view is replaced by another'.
- 3 The formulation of scientific method outlined earlier has come in for strong and sustained criticism. Mishler (1990), for example, describes it as a 'storybook image of science', out of tune with the actual practices of working scientists who turn out to resemble craftspersons rather than logicians. By craftspersons, Mishler is at pains to stress that competence depends upon 'apprenticeship training, continued practice and experienced-based, contextual knowledge of the specific methods applicable to a phenomenon of interest rather than an abstract "logic of discovery" and application of formal "rules"'. The knowledge base of scientific research, Mishler contends, is largely tacit and unexplicated; moreover, scientists learn it through a process of socialization into a 'particular form of life'. The discovery, testing and validation of findings is embedded in cultural and linguistic practices and experimental scientists proceed in pragmatic ways, learning from their errors and failures, adapting procedures to their local contexts, making decisions on the basis of their accumulated experiences. See, for example, Mishler (1990).
- 4 Investigating social episodes involves analysing the accounts of what is happening from the points of view of the actors and the participant spectator(s)/investigator(s). This is said to yield three main kinds of interlocking material: images of the self and others, definitions of situations, and rules for the proper development of the action. See Harré (1976).



Companion Website

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